

Maximum-Likelihood Learning of Latent Dynamics Without Reconstruction

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Abstract

We address the challenge of uncovering systematic, and potentially controllable, dynamical structure underlying complex high-dimensional time-series data. Existing generative and autoregressive models have difficulty telling systematic content apart from distractors, while contrastive methods struggle to learn accurate dynamics. To address these shortcomings, we develop the Recognition-Parametrized Gaussian State-Space Model (RP-GSSM), a probabilistic framework that infers accurate latent dynamics without relying on a parametrized decoder. By eliminating explicit generative parameters, the model directs its entire representational capacity to encoding dynamically relevant state; and, being fully probabilistic, learns via maximum likelihood without auxiliary objectives or ad-hoc regularization. Combining the expressive power of a neural network encoder with exact inference under a jointly Gaussian prior allows the RP-GSSM to embed a broad class of intrinsically nonlinear dynamical systems. The RP-GSSM parameters are provably linearly identifiable in the infinite data limit. We show that the RP-GSSM recovers physically meaningful latent states from noisy video more faithfully than competing methods, more reliably identifies underlying controllable nonlinear dynamics, and remains substantially more robust to visual distractors.

1. Introduction

Input observations to embodied intelligence, whether biological or machine-based, naturally arrive in time series. A key goal for such intelligence is to extract a dynamically predictable and actionable state variable from those inputs. This world state may be needed to plan control

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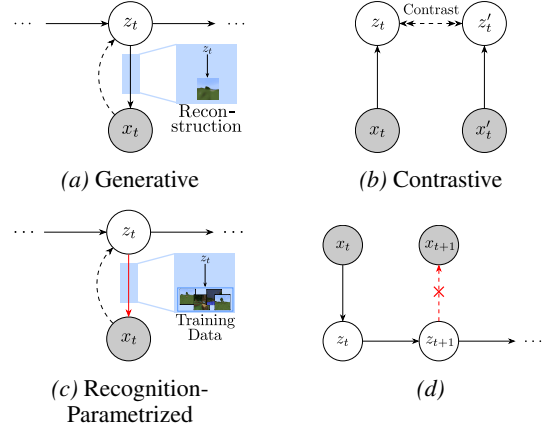


Figure 1. (a-c): Different approaches to modeling latent variables z_t underlying observed time-series data x_t ; (d): In many applications, future latent variables are predicted, but not future observations.

strategies, make other decisions, or perform ancillary predictions in downstream tasks. Here we focus on the fundamental problem of inferring this state and its dynamics from high-dimensional time-series data.

Most existing models fall into two broad categories: *generative* or *contrastive* (Liu et al., 2023; Zhang et al., 2024).

Generative methods parametrize a joint distribution over latent and observed variables, and aim to maximize the data likelihood. Crucially, the joint distribution is defined using an explicit likelihood that models how observations are generated from latents. The most common examples are dynamical variants of the variational autoencoder (VAE; Kingma & Welling, 2014; Gregor et al., 2015; Johnson et al., 2016). VAEs are trained using approximate maximum likelihood estimation, enjoying convergence guarantees (subject to approximation error) and principled uncertainty estimation. They work by learning a variational posterior parametrized by a recognition network (or “encoder”), which is trained concurrently with the generative network (or “decoder”; Figure 1a). Thus, a trained VAE can perform both recognition and generation. However, in many problem domains, such as decision making and control, agents require prediction of future latent states but *not* future observations (Figure 1d). In this case, training a generative network is computationally wasteful and can

be detrimental to learning (Balestrierio & LeCun, 2024). Indeed, VAE-based losses can be interpreted as regularized reconstruction objectives, which encourage latents to encode information about all aspects of the data, including irrelevant distractor features. Further, flexible decoders may unsuccessfully attempt to compensate for errors in encoders (Cremer et al., 2018), leading to biased parameter estimates and poor representations (Turner & Sahani, 2011).

Contrastive methods, on the other hand, learn latent representations without modeling how observations are generated. Instead, they train a recognition network to contrast “positive” observation pairs from “negative” ones (Figure 1b). Contrastive methods do not learn a full probabilistic state-space model, therefore lacking natural notions of uncertainty and predictive latent dynamics, and risking model collapse. Failure to model uncertainty can also lead to poorer representations, particularly in time-series models (Welch & Bishop, 1995).

We aim to improve recognition of latent dynamics by combining the best of both worlds: utilizing the likelihood formalism of probabilistic models, but without an explicit generative decoder. We do this in the framework of recognition-parametrized models (RPMs; Walker et al., 2023). The decoder in an RPM is defined implicitly by the recognition process, allowing the model to focus capacity on learning a good encoder (Figure 1c). Furthermore, the RPM likelihood emphasizes latent structure that captures the joint distribution over multiple observations rather than accurate reconstruction of any single observed variable. In time series, this amounts to capturing information that systematically accounts for correlations and regularities amongst observations at different times.

The organization of the paper is as follows. In Section 2 we review essential background on RPMs and state-space models. In Section 3, we combine the two, extending the RPM framework to the case of latent dynamical systems. We also provide a linear identifiability result. Section 4 places our work in context. Finally, in Section 5 we demonstrate on a variety of datasets the ability of the RP-GSSM to learn good latent representations from high-dimensional time-series observations. We also show how the proposed approach systematically overcomes basic failure modes of a wide range of baseline methods, both generative and contrastive, for unsupervised representation learning for time series.

2. Background

2.1. Recognition-Parametrized Models

We first introduce the RPM in its general form, following Walker et al. (2023). Consider a latent variable model

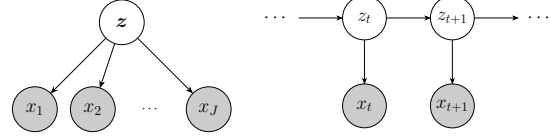


Figure 2. **Left:** Latent variable model with multiple observation factors; **Right:** The SSM graphical model.

where the observation $\mathbf{x} := (x_1, \dots, x_J)$ comprises multiple variables (i.e., $J > 1$) that are conditionally independent given the latents $\mathbf{z}$ (Figure 2, left):

$$x_i \perp\!\!\!\perp x_j \mid \mathbf{z} \quad \forall i \neq j.$$

The boldface $\mathbf{z}$ may represent multiple latent variables with internal conditional independence structure, such as a Markov chain, as we will see later. The graphical model implies a joint distribution that factorizes as

$$p(\mathbf{x}, \mathbf{z}) = p(\mathbf{z}) \prod_{j=1}^J p(x_j \mid \mathbf{z}).$$

Equivalently, by Bayes’ rule,

$$p(\mathbf{x}, \mathbf{z}) = p(\mathbf{z}) \prod_{j=1}^J \frac{p(\mathbf{z} \mid x_j) p(x_j)}{p(\mathbf{z})}.$$

Given a dataset $\mathbf{X} = (x_1^n, \dots, x_J^n)_{n=1}^N$, Walker et al. (2023) propose to model the true $p(\mathbf{x}, \mathbf{z})$ with a semi-parametric distribution $\tilde{p}_\theta^{\mathbf{X}}(\mathbf{x}, \mathbf{z})$ called the *RPM joint*, given by

$$\tilde{p}_\theta^{\mathbf{X}}(\mathbf{x}, \mathbf{z}) = p_\eta(\mathbf{z}) \prod_{j=1}^J \frac{f_{\phi_j}(\mathbf{z} \mid x_j) p_{0j}(x_j)}{F_{\phi_j}(\mathbf{z})}, \quad (1)$$

where $\theta = (\eta, \{\phi_j\}_{j=1}^J)$ and

- p_η is a parametrized prior;
- $f_{\phi_j}(\mathbf{z} \mid x_j)$ are parametrized x_j -dependent distributions over $\mathbf{z}$ called *recognition factors*;
- p_{0j} is the empirical data distribution on the j th observation modality:

$$p_{0j}(x) = \frac{1}{N} \sum_{n=1}^N \delta(x = x_j^n);$$

- $F_{\phi_j}(\mathbf{z})$ ensures that the RPM joint is normalized:

$$\begin{aligned} F_{\phi_j}(\mathbf{z}) &= \int f_{\phi_j}(\mathbf{z} \mid x_j) p_{0j}(x_j) dx_j \\ &= \frac{1}{N} \sum_{n=1}^N f_{\phi_j}(\mathbf{z} \mid x_j^n). \end{aligned} \quad (2)$$

The superscript $\mathbf{X}$ in the RPM joint emphasizes that it is itself a function of the training data through its dependence on the empirical distributions $p_{0j}(x_j)$.

Equation (1) defines a joint distribution over observed and latent variables that avoids parametrizing the generative process $p(\mathbf{x}|\mathbf{z})$. In fact, the generative distribution implied by the RPM joint in Equation (1) is

$$\tilde{p}_{\theta}^{\mathbf{X}}(\mathbf{x}|\mathbf{z}) \propto \prod_{j=1}^J f_{\phi_j}(\mathbf{z}|x_j) p_{0j}(x_j),$$

the support of which is constrained to $\prod_j p_{0j}(x_j)$. Hence, as a *generative* model, the RPM with atomic p_{0j} is formally limited to only generate values of x_j that appear in the training set. Yet, the recognition factors themselves are parametric, and so they support *inference* for previously unseen observations.

The RPM joint is a normalized distribution and can be fit by maximum likelihood using the Expectation Maximization (EM) algorithm and related methods (Dempster et al., 1977). Walker et al. (2023) implement RPMs for a variety of different data sets and priors, introducing various approximations to the likelihood when necessary. We apply and extend some of their methods in Section 3.

2.2. State-Space Models

A probabilistic SSM¹ is a latent variable model in which observations $\mathbf{x} := (x_1, \dots, x_T)$, $x_t \in \mathbb{R}^{\mathcal{D}_x}$ are assumed to be independently generated given a *latent* time series $\mathbf{z} := (z_1, \dots, z_T)$, $z_t \in \mathbb{R}^{\mathcal{D}_z}$ that evolves according to Markovian dynamics:

$$p(\mathbf{x}, \mathbf{z}) = p(z_1) \prod_{t=2}^T p(z_t|z_{t-1}) \prod_{t=1}^T p(x_t|z_t). \quad (3)$$

This structure corresponds to the graphical model shown in Figure 2 (right). Note that an SSM is a special case of the general latent-conditioned model discussed in Section 2.1.

The standard approach to fitting an SSM to data is to parametrize the terms of Equation (3) with parameters θ , define a *variational posterior* $q(\mathbf{z}) \approx p_{\theta}(\mathbf{z}|\mathbf{x})$, and maximize the *free energy* (Neal & Hinton, 1998)

$$\mathcal{F}(q, \theta) = \langle \log p_{\theta}(\mathbf{x}|\mathbf{z}) \rangle_{q(\mathbf{z})} - \text{KL}(q(\mathbf{z}) || p_{\theta}(\mathbf{z})), \quad (4)$$

where $\langle \cdot \rangle_{q(\mathbf{z})}$ denotes expectation with respect to q . The free energy is a lower bound to the data log-likelihood, $\log p_{\theta}(\mathbf{x})$, and can be maximized via the EM algorithm,

¹We use the term “SSM” to refer to a generic probabilistic model with graphical structure as in Figure 2 (right), not the deterministic methods in recent literature on structured state-space models, e.g. Mamba (Gu & Dao, 2024).

consisting of alternating updates to q (“E-step”) and θ (“M-step”) (Dempster et al., 1977). Although the EM algorithm can be performed exactly for discrete latent variables (Baum & Petrie, 1966)—albeit with exponential computational complexity in latent dimensionality—in this work we focus on continuous latent variables. In this setting, there are few models for which EM can be performed exactly.

2.2.1. GAUSSIAN SSM

In a *Gaussian SSM* (GSSM), the components of the prior distribution over the latents take the following jointly Gaussian form:

$$p(z_1) = \mathcal{N}(m_1, Q_1), \quad p(z_t|z_{t-1}) = \mathcal{N}(Az_{t-1} + b, Q).$$

where $p(z_1)$ is an *initial distribution*, and $p(z_t|z_{t-1})$ is the *transition distribution*.

If the *emission distribution* is also linear-Gaussian, $p(x_t|z_t) = \mathcal{N}(Cz_t + d, R)$, then both E- and M-steps are fully tractable: the model admits exact inference by Kalman smoothing (Kalman, 1960), and the M-step can be performed in closed-form (Ghahramani & Hinton, 1996). However, exact inference is not possible with nonlinear emissions, and linear emissions are overly restrictive for many real-world datasets.

Our focus here is on models where the link between z_t and x_t is nonlinear and flexibly parametrized. See Section 3.4 for a discussion of the implications of joint Gaussianity in the latents.

2.2.2. STABLE GAUSSIAN SSMS AND STANDARDIZED PARAMETRIZATION

A GSSM is *stable* if $\rho(A) < 1$, where $\rho(\cdot)$ denotes the largest eigenvalue magnitude. A stable system has a well-defined *stationary state distribution* $\lim_{t \rightarrow \infty} p(z_t)$ (Buesing et al., 2012). Stable GSSMs are unidentifiable: the following lemma shows that without loss of generality, any stable GSSM can be transformed to one with zero transition bias and a $\mathcal{N}(0, I)$ stationary state distribution without altering the distribution of observations.

Lemma 2.1. *Let the GSSM on observations $\mathbf{x} = (x_1, \dots, x_T)$ and latents $\mathbf{z} = (z_1, \dots, z_T)$, with parameters $\Theta = (m_1, Q_1, A, b, Q)$ and an (arbitrary) emission distribution $e(x_t|z_t)$, be stable. Then there exists another stable GSSM with parameters $\tilde{\Theta} = (\tilde{m}_1, \tilde{Q}_1, \tilde{A}, \tilde{b}, \tilde{Q})$ and emission distribution $\tilde{e}$ such that $p_{\Theta, e}(\mathbf{x}) = p_{\tilde{\Theta}, \tilde{e}}(\mathbf{x})$, the stationary distribution of the latent variables (under $\tilde{\Theta}$) is $\mathcal{N}(0, I)$, and $\tilde{b} = 0$.*

We provide a proof of Lemma 2.1 and an explicit form of $\tilde{\Theta}$ and $\tilde{e}$ in Appendix A.

3. RP-GSSM

In this section we apply the RPM framework to the GSSM and develop the necessary approximations for learning and inference. The RPM joint for the SSM graphical model is

$$\hat{p}_\theta^{\mathbf{X}}(\mathbf{x}, \mathbf{z}) = p_\eta(z_1) \prod_{t=2}^T p_\eta(z_t | z_{t-1}) \prod_{t=1}^T \frac{f_{\phi_t}(z_t | x_t) p_{0t}(x_t)}{F_{\phi_t}(z_t)}. \quad (5)$$

Inspired by the temporal factorization of emissions in the SSM joint (Equation 3), we restrict the recognition factor distributions to the single latent state at the time of the observation:

$$\hat{p}_\theta^{\mathbf{X}}(\mathbf{x}, \mathbf{z}) = p_\eta(z_1) \prod_{t=2}^T p_\eta(z_t | z_{t-1}) \prod_{t=1}^T \frac{f_{\phi_t}(z_t | x_t) p_0(x_t)}{F_{\phi_t}(z_t)}. \quad (6)$$

For tractability, we choose a stable GSSM prior with the parametrization from Lemma 2.1 and neural network-Gaussian recognition factors:

$$p(z_1) = \mathcal{N}(0, I), \quad p_\eta(z_t | z_{t-1}) = \mathcal{N}(Az_{t-1}, Q), \quad (7)$$

$$f_{\phi_t}(z_t | x_t) = \mathcal{N}(\mu_{\phi_t}(x_t), \Sigma_{\phi_t}(x_t)).$$

We refer to this form as a recognition-parametrized GSSM (RP-GSSM). We will see later that this setup ensures that the posterior $\hat{p}_\theta^{\mathbf{X}}(\mathbf{z} | \mathbf{x})$ is Gaussian and can be computed exactly in closed form, despite the arbitrary neural network link between observations and latents.

3.1. The RP-GSSM Free Energy

For simplicity, we present the free energy for a single data sequence $\mathbf{x} = (x_1, \dots, x_T)$; these may be summed over sequences when given a batch of time-series observations. Combining Equations (4) and (6) we have

$$\mathcal{F}(q, \theta) = \sum_{t=1}^T \left(\left\langle \log \frac{f_{\phi_t}(z_t | x_t)}{F_{\phi_t}(z_t)} \right\rangle_{q(z_t)} + \log p_{0t}(x_t^n) \right) - \text{KL}(q(\mathbf{z}) || p_\eta(\mathbf{z})). \quad (8)$$

The $\log p_{0t}$ terms do not depend on θ or q , so we omit them below. Note that $F_{\phi_t}(z_t)$ is a mixture of Gaussians (implied by Equations 2 and 7) and so $\langle \log F_{\phi_t}(z_t) \rangle_{q(z_t)}$ is intractable. Rather than resorting to sample-based Monte Carlo approximations, we resolve this term by adopting the *interior variational bound* (Walker et al., 2023). Introducing parametrized auxiliary factors $g_\omega^t(z_t)$, which may be unnormalized, and applying Jensen’s inequality, we obtain

$$-\langle \log F_{\phi_t}(z_t) \rangle_{q(z_t)} = -\left\langle \log \frac{F_{\phi_t}(z_t) g_\omega^t(z_t) q(z_t)}{g_\omega^t(z_t) q(z_t)} \right\rangle_{q(z_t)} \geq \left\langle \log \frac{g_\omega^t(z_t)}{q(z_t)} \right\rangle_{q(z_t)} - \log \int F_{\phi_t}(z_t) g_\omega^t(z_t) dz_t. \quad (9)$$

We choose a parametrization of the g_ω^t and f_{ϕ_t} that limits parameter count, simplifying optimization, while keeping the model expressive. Note that the Jensen bound in Equation (9) is tight when $g_\omega^t(z_t) \propto q(z_t)/F_{\phi_t}(z_t)$. As discussed by Walker et al. (2023), $F_{\phi_t}(z_t) \rightarrow p_\eta(z_t)$ in the large-data in-model limit. To support this convergence, we assume $F_{\phi_t}(z_t) \approx p_\eta(z_t)$ and set $g^t(z_t) = q(z_t)/p_\eta(z_t)$, dropping the ω subscript as there are no additional parameters to optimize. Appendix F provides empirical evidence that the approximation $F_{\phi_t}(z_t) \approx p_\eta(z_t)$ holds in practice.

Next, note that the true marginal posterior $p(z_t | x_t)$, which the recognition factors $f_{\phi_t}(z_t | x_t)$ aim to approximate, is proportional to $p(z_t)p(x_t | z_t)$ by Bayes’ rule. Conditioned on z_t , the generative mapping $z_t \mapsto x_t$ is independent of t . Therefore, we decompose the recognition factors into a time-varying prior term and a time-invariant recognition term: $f_{\phi_t}(z_t | x_t) \propto p_\eta(z_t) f_\phi^\Delta(z_t | x_t)$, where the f_ϕ^Δ are learned time-invariant distributions: $f_\phi^\Delta(z_t | x_t) = \mathcal{N}(\mu_\phi(x_t), \Sigma_\phi(x_t))$.

The full parameter set is now $\theta = (\eta, \phi)$, for $\eta = (A, Q)$. Applying the lower bound in Equation (9) yields a lower bound to $\mathcal{F}$, which we call the *auxiliary free energy* and denote $\mathcal{G}(q, \theta)$:

$$\mathcal{G}(q, \theta) \stackrel{+c}{=} \sum_{t=1}^T \left(\langle \log f_\phi^\Delta(z_t | x_t) \rangle_{q(z_t)} - \log \Gamma_\phi^t \right) - \text{KL}(q(\mathbf{z}) || p_\eta(\mathbf{z})), \quad (10)$$

where the $\log \Gamma_\phi^t$ are “log-sum-exp-like” terms which can be computed in closed form (Appendix C).

3.2. Training the RP-GSSM

The RP-GSSM is trained by maximizing $\mathcal{G}$ via alternating updates to q and θ , as described in Algorithm 1. In practice, we sum over a batch of data sequences at each iteration.

Despite its nonlinear link between observations and latents, the RP-GSSM’s update to q given fixed θ is closed form. Up to an additive constant independent of q , $\mathcal{G}$ has the same form as the free energy of a GSSM (Equation 4) with emission factor $p(x_t | z_t)$ replaced by $f_\phi^\Delta(z_t | x_t)$. A GSSM with $p(x_t | z_t) = \mathcal{N}(Cz_t + d, R)$ would admit an optimal q that is Gaussian and can be computed via Kalman smoothing. The optimal RP-GSSM q is therefore also Gaussian and is computed by Kalman smoothing with C, d, R , and data x_t replaced by $I, 0, \Sigma_\phi(x_t)$, and $\mu_\phi(x_t)$. We call this procedure “augmented Kalman smoothing” and provide a full description in Appendix C.3. The ability to perform closed-form inference means the RP-GSSM avoids Monte Carlo approximations, ensuring graceful scaling with latent dimensionality, as we show in Section 5.1.

Given Gaussian q , $\mathcal{G}$ can be computed in closed form and

Algorithm 1 RP-GSSM Training

Input: Initial parameters $\theta = \{A, Q, \phi\}$, data $x_{1:T}$, learning rate η
Output: Optimized parameters θ^*
while not converged **do**
 // E-Step
 $q \leftarrow \text{AugmentedKalmanSmoother}(\theta, x_{1:T})$
 // M-Step
 Compute $\mathcal{G}(q, \theta)$ as in Equation (10)
 $\theta \leftarrow \theta + \eta \nabla_{\theta} \mathcal{G}(q, \theta)$
end while
return θ

θ is updated by gradient ascent. To ensure that the prior remains stable, after each gradient step we project A onto the set of matrices with singular values < 1 (which implies $\rho(A) < 1$) (Buesing et al., 2012; Zhao & Linderman, 2023). A full derivation of $\mathcal{G}$ and a discussion of the variational gap introduced by the interior variational bound are provided in Appendix C.

3.3. Identifiability

We show that the model of Equation (5) is linearly identifiable in the limit of infinite data. By the Glivenko-Cantelli theorem, each empirical distribution $p_{0t}(x_t)$ converges uniformly to the true data distribution $p_t(x_t)$ as the number of i.i.d. data sequences increases. We refer to the resulting model, which the RP-GSSM approximates, as a *limiting RP-GSSM* and denote it $\tilde{p}_{\theta}$.

Theorem 3.1. *Let $\tilde{p}_{\theta}$ and $\tilde{p}_{\theta'}$ be two limiting RP-GSSMs with parameters $\theta = (A, Q, \eta)$, $\theta' = (A', Q', \eta')$, where η denotes a collection of arbitrarily nonstationary recognition factor natural parameters: $(\eta_1, \dots, \eta_T)$. Assume that*

- (i) *Each η_t has open image;*
- (ii) *The η_t are differentiable with full-rank Jacobians almost everywhere;*
- (iii) *Q and Q' are positive definite;*
- (iv) *$\exists L > 0$ such that $\nabla^2 \log F_t(z) \preceq LI \forall t, z$;*
- (v) *$T \geq 3$.*

If $\tilde{p}_{\theta}(x) = \tilde{p}_{\theta'}(x)$, then there exists an invertible linear mapping relating the models' natural parameters and latent spaces: $z = Gz'$.

A proof is provided in Appendix B.

3.4. Limitations

The primary limitation of the RP-GSSM is its linear-Gaussian prior. However, this is ameliorated by the

flexible nonlinear link between observations and latents. Koopman operator theory represents nonlinear dynamics through a linear operator acting on observables, motivating eigenspace embeddings in which nonlinear dynamics may be represented, exactly or approximately, by linear evolution (Koopman, 1931; Brunton et al., 2022). The RP-GSSM recognition factors can be thought of as learning such a Koopman eigenspace. The ambient dimension necessary for the embedding may be larger than the intrinsic dimensionality of the system, but often remains lower than that of the observations. In systems with transition noise or a continuous spectrum of oscillatory frequencies (such as a pendulum), the Koopman embedding becomes approximate or infinite-dimensional, respectively. We include examples of such systems in Section 5 to illustrate that the RP-GSSM is able to handle these cases in practice.

Another potential limitation of the RP-GSSM is the lack of explicit decoder, which means that it is unable to generate novel observations. While we have stressed that generative reconstruction in observation space is a suboptimal objective for learning latent representations, generation is sometimes useful for visualization or for downstream tasks. Even though the RP-GSSM avoids the liabilities associated with explicit reconstruction during learning and inference, it remains possible to learn a separate generative decoder from the learned latent space. Such a *post-hoc* decoder can be trained by simple regression from the learned latent representation (often the posterior mean) to the corresponding observed variable, holding the recognition model fixed. Thus, the auxiliary decoder does not alter the RP-GSSM latent state in any way. Section 5 shows that *post-hoc* generation from the RP-GSSM can filter out “unnecessary” elements (noise, distractions, etc.) in the observations better than trained generative models.

Finally, the approximations necessary to obtain a closed-form loss, detailed in Section 3.1, may bias the RP-GSSM towards suboptimal solutions. While the approximations are often tight in practice (Appendix F), a detailed study of their implications is important future work.

4. Related Work

Early extensions of Kalman smoothing to nonlinear transition and emissions distributions include the Extended Kalman filter (EKF; Jazwinski, 1970), which linearizes around posterior means, and the Unscented Kalman filter (Wan & Van Der Merwe, 2000), which uses a quadrature-based approach to extend approximation up to third order.

One of the first successful VAE-based time-series models was the Deep Kalman Filter (DKF; Krishnan et al., 2015). The DKF parametrizes the variational posterior q with a recurrent neural network (RNN) and uses reparametrization

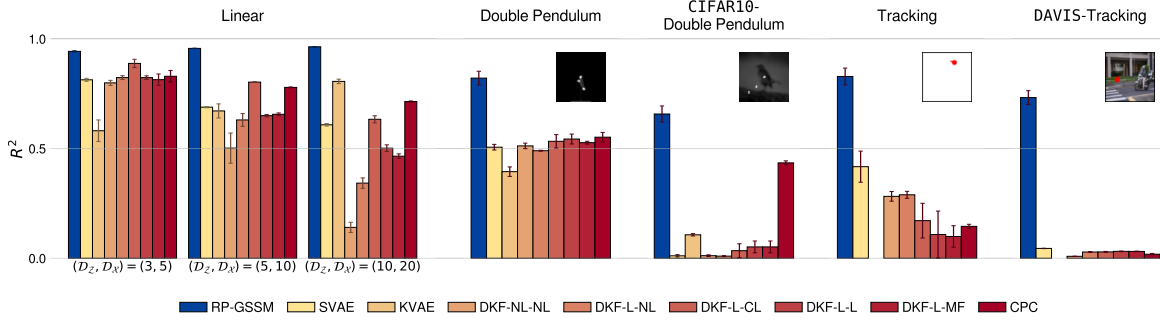


Figure 3. Validation linear regression R^2 scores between inferred posterior means and ground truth latent state variables across multiple datasets. Error bars indicate standard error over three seeds. We find the RP-GSSM learns more informative latent representations than alternatives in both the absence and the presence of background distractors.

to compute gradients through a sample-estimated free energy. The DKF forms the foundation of PlaNet (Hafner et al., 2019) and DreamerV3 (Hafner et al., 2023), a highly successful reinforcement learning framework. The Kalman VAE (KVAE; Fraccaro et al., 2017) and Recurrent Kalman Network (RKN; Becker et al., 2019), on the other hand, combine *locally* linear latent dynamics with VAE-based generative networks. The KVAE exploits local linearity to enable Kalman smoothing-based posterior inference, whereas the RKN combines local linearity with factorized state beliefs to learn high-dimensional latent state representations efficiently. The KVAE comprises the backbone of the recent K^2 VAE (Wu et al., 2025).

Perhaps the generative method closest to the RP-GSSM is the structured VAE (SVAE; Johnson et al., 2016; Zhao & Linderman, 2023), which learns a recognition network that outputs conjugate potentials on latents. When paired with linear-Gaussian transitions, the potentials allow for exact inference via Kalman smoothing. However, unlike the RP-GSSM, the SVAE also learns an explicit generative model using reparametrization.

Contrastive Predictive Coding (CPC) (van den Oord et al., 2018) exploits the contrastive InfoNCE loss to identify a latent representation that maximizes internal predictability. Here, a recognition network summarizing past data aims to discriminate future latents from the same sequence (positive) from draws taken from other sequences (negative). Models that build on CPC include Contrastive RL (Eysenbach et al., 2022) and TS-TCC (Eldele et al., 2021). Recent extensions of contrastive learning seek to learn explicit latent dynamics by adding an auxiliary loss (You et al., 2022) or by integrating parametrized dynamics directly into the similarity function (Laiz et al., 2025).

5. Experiments

We evaluated the RP-GSSM against generative and contrastive approaches on tasks of increasing complexity and

dimensionality, focusing on challenging settings where the observations included distractors such as background images or videos, observation noise, and changes in camera angle. The reconstruction-based objectives of generative models encouraged reconstruction of the irrelevant distractors, impeding latent state inference. The RP-GSSM instead focused on learning optimal recognition and latent dynamics models. We found that the RP-GSSM matched or outperformed baselines when recovering accurate latent representations across all tasks, especially when distractors are present in the observed data.

We used CPC, SVAE, KVAE, and DKF with five different parametrizations as benchmarks. The DKF parametrizations (detailed in Appendix D) were chosen to explore the effects of linear or MLP prior transitions, and various posterior approximations: mean-field over time, linear, or non-linear with either RNN or temporal CNN recognition.

As a proxy for performance across a wide range of downstream applications, we scored each model with the linear regression R^2 between inferred latent trajectories and ground-truth state trajectories, across a batch of validation sequences. We used best-practice training for the baseline models where appropriate, such as self-supervised masking for the generative methods (Zhao & Linderman, 2023) and LayerNorm for stability of nonlinear priors (Ba et al., 2016). Model architectures were kept as similar as possible. Each dataset comprised sequences of equal length and each model was trained using batches of full-length sequences. We report the best performance of each model from a hyperparameter search over weight initialization, latent dimension, diagonal vs. full recognition covariance parametrization, and magnitude of decoder covariance (where applicable). A full description of hyperparameters, datasets, and results with nonlinear regression is detailed in Appendices D, E, and F, respectively. We provide animated visualizations at rpgssm.github.io.

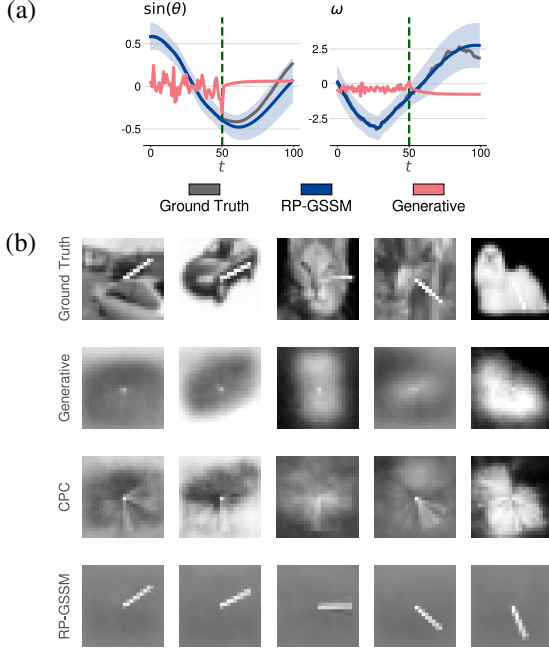


Figure 4. (a) Predictions of true states on CIFAR10-Pendulum, given 50 frames of context. (b) Reconstructions on CIFAR10-Pendulum for the best generative baseline and auxiliary decoders trained using the best CPC and RP-GSSM models.

5.1. Linear Dynamical System with Linear Emissions

The first evaluation was on a fully in-model-class dataset: samples from a GSSM with linear-Gaussian emissions. Even in this simple setting, the RP-GSSM outperformed competing methods, particularly for larger dimensionalities (Figure 3, left).

5.2. Pendulum and Double Pendulum

Next, we considered video data from systems with nonlinear ground truth dynamics: noisy synthetic video of a single pendulum (Becker et al., 2019), and real video of a double pendulum (Asseman et al., 2018), both evolving freely under gravity from random initial conditions. Recognition and generative networks were all three-layer convolutional and deconvolutional neural networks, respectively. The true underlying dynamics of a pendulum form a nonlinear dynamical system governed by angle and angular velocity. We used these two variables for our regression targets. The dynamics of a double pendulum are higher-dimensional, nonlinear, and chaotic. In this case our targets were the angle of the primary pendulum relative to the origin and the angle of the secondary pendulum relative to the primary pendulum. The mean R^2 scores to all targets are shown in Figure 3 for the double pendulum, and in Appendix F for pendulum. All methods performed reasonably well at decoding the targets, with the RP-GSSM being most consistent. In Appendix F we provide further results

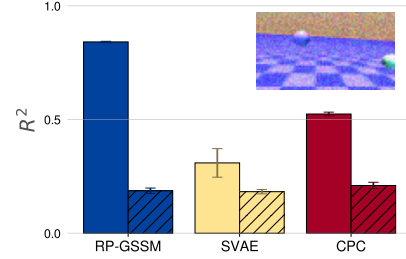


Figure 5. Validation performance on the distracting egocentric task. Solid and hatched bars indicate kernel ridge regression and linear regression R^2 , respectively. Mean R^2 is computed over three seeds, with error bars indicating standard error. Sample frame shown top right.

demonstrating the increased data efficiency of RP-GSSM compared to baseline methods.

5.2.1. CIFAR10 BACKGROUND DISTRACTIONS

To evaluate the impact of distractors, we superimposed the pendulum image sequences over images from the CIFAR10 collection (Krizhevsky & Hinton, 2009) drawn randomly at each time step (CIFAR10-Pendulum and CIFAR10-Double Pendulum). Regression R^2 results on these data are shown in Figure 3. It is evident that only the RP-GSSM captured the underlying dynamical system reliably in the presence of distractors.

The benefit of modeling only dynamically structured features was evident in predictions of future latent states. Figure 4a shows forecast future latent states from the RP-GSSM and the best-performing generative alternative.

We investigated the consequences of these latent-state differences on encoded observation-domain features by training auxiliary decoders to map the latents inferred by the best RP-GSSM and CPC runs on CIFAR10-Pendulum to the corresponding images. The decoders recapitulated the architecture of the generative baseline models, and were trained with the same objective: $\max_{\theta} \langle \log p_{\theta}(z|x) \rangle_{q(z)}$. Full details appear in Appendix D.

Figure 4b compares reconstructions from the RP-GSSM and CPC with those of the best performing generative model. The distinction between model representations is striking. The generative model attempts to reconstruct each image pixel by pixel, resulting in a blurred reconstruction of the background and absence of the pendulum arm. CPC’s encoding is mixed: while it contains some information about the pendulum, a lack of coherent dynamics model means that the full pendulum dynamics are lost. On the other hand, the RP-GSSM only encodes the temporally persistent pendulum, while ignoring information about the background images, indicated by the grey reconstruction backgrounds (an average over all possible backgrounds).

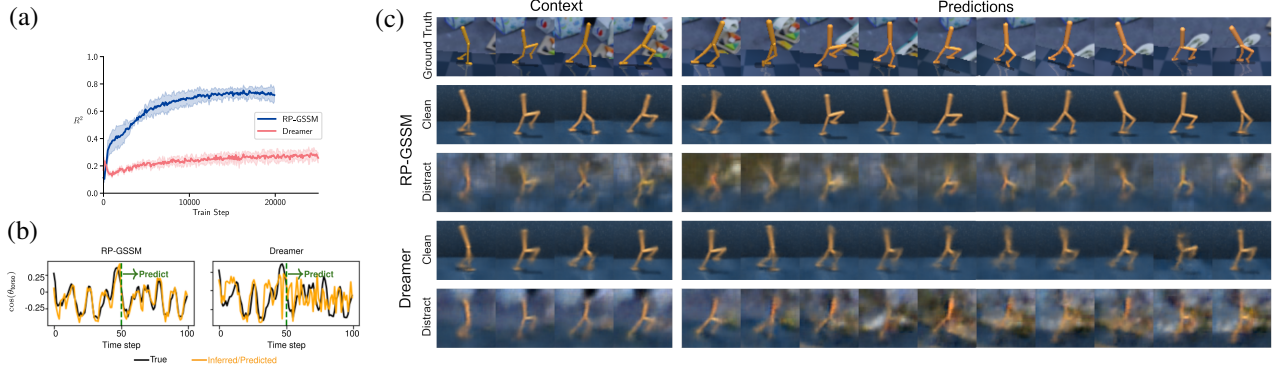


Figure 6. (a) Validation kernel ridge regression R^2 scores over training. (b) Predictions of cosine of torso angle, given 50 frames of context. (c) Reconstructions and predictions of the auxiliary decoders on a held-out trajectory. **Context** frames are reconstructed from inferred posterior means, and **Prediction** frames are generated by sampling $z_{\tau+1:T} \sim p(z_{\tau+1:T} | x_{1:\tau}, u_{1:T-1})$, where τ is context length, and then applying the decoder.

5.3. Object State Tracking

We applied the RP-GSSM to object state tracking from video. This task is of particular interest in real-world applications such as robotics and behavioral neuroscience. The data comprised synthetic video sequences simulating an agent running in a rectangular arena (George et al., 2024). The agent is rendered as a red dot with a head direction arrow. The true dynamics are governed by an Ornstein-Uhlenbeck (OU) process on the velocity vector. The walls of the arena enforce hard nonlinearities on agent dynamics. Regression targets were six-dimensional, consisting of position, velocity, and head direction vectors. Although free-field latent dynamics are linear, the walls introduce local discontinuities. Models were fit to high-dimensional image observations (Figure 3).

5.3.1. DAVIS BACKGROUND DISTRACTIONS

We introduced temporally correlated distractors to the tracking data by incorporating randomly selected 2s-long video clips from the DAVIS dataset (Perazzi et al., 2016; Pont-Tuset et al., 2017) in the background (DAVIS-Tracking). Although the backgrounds have their own temporally consistent dynamics, different data sequences have different background videos, and so a good model should learn to track features that are consistent across the entire dataset, here the movement of the agent.

The mean R^2 scores for agent-state regression are shown in the final column of Figure 3. The RP-GSSM was the only algorithm able to consistently track the agent. Further results investigating the RP-GSSM’s robustness to changes in background distractor on the tracking task are shown in Appendix F.

5.4. First-Person Visual Input

We simulated first-person visual motion input by mapping marker-based human motion-capture data from running sequences in the LAFAN1 dataset (Harvey et al., 2020) to a biomechanically-detailed musculoskeletal simulation (LoCoMuJoCo; Al-Hafez et al., 2023) with an egocentric camera rigidly attached to the skull body of the skeleton. This provided a first-person view that moved with the simulated body, rendered at 30 Hz in 80×45 RGB frames. The ground truth states were 7-dimensional, comprising 3D position and orientation parametrized as unit quaternions.

Five classes of visual distractor were superimposed to confuse low-level visual cues tied to the body’s pose or environment: random floor tile color, random ambient lighting level, atmospheric fog of random color and density, i.i.d. pixel noise, and three moving spheres. The moving spheres had random colors and moved according to an OU process with stationary distribution centered in front of the agent to ensure significant visual overlap.

Figure 5 shows mean validation R^2 from kernel ridge regression and linear regression, comparing the RP-GSSM to SVAE (the best-performing generative model on average) and CPC. All models were trained with 7-dimensional latents and provided comparable performance with linear regression. The RP-GSSM significantly outperformed the baselines with kernel ridge regression.

5.5. Nonlinear Controlled Dynamics

Finally, we considered a system with nonlinear dynamics, nonlinear control inputs, and significant visual clutter: simulations of the Walker environment of the DeepMind Control Suite (Tunyasuvunakool et al., 2020), with

added background videos, random camera pose changes, and randomized agent colors, generated by [Ortiz et al. \(2024\)](#) under a fixed control policy. The internal system state is 24-dimensional, comprising the agent’s height and the velocities and orientations of its joints, although these variables may be tightly correlated under well-trained policies. The control signal is the torque applied to each of the agent’s six joints. To incorporate control information into the RP-GSSM, we conditioned the latent transition on the control inputs u_t , modeling the prior as $p(z|u) = p(z_1) \prod_{t=1}^T p(z_t|z_{t-1}, u_{t-1})$ with the transition matrix determined by a neural-network function of the control input: $A_\psi(u_{t-1})$. Despite its nonlinear dependence on $\{u_t\}_t$, the prior remains linear-Gaussian in the latents, and so inference could still be performed exactly by Kalman smoothing.

We compared the RP-GSSM to the world model of Dreamer ([Hafner et al., 2020](#)), a model-based reinforcement method. Both models had 8-dimensional latents, which we found sufficient to capture the latent dynamics due to policy-induced state correlations. The RP-GSSM learned to extract state information (Figure 6a) and predict dynamics (Figure 6b) more rapidly and accurately. To predict torso angle, the RP-GSSM must learn the indirect effect of limb-joint control on the torso through complex gravity- and contact-dependent dynamics. Rollouts of all latent states are provided in Appendix F. We also trained auxiliary decoders to both the original images as well as “clean” images without distractor elements, and visualized rollouts from both models (Figure 6c). The RP-GSSM representations encode the agent configuration more reliably than Dreamer, and the learned dynamics model is able to predict future states from control inputs accurately for at least 50 timesteps. The greater accuracy of “clean” decoding reflects the preference for the learned state to reflect dynamically predictable elements. This difference was more pronounced for the RP-GSSM, exemplifying the advantage of learning latent state without the need to regenerate detailed pixel-level data. Appendices E and F contain further details and results of models trained with 32-dimensional latents.

Figure 7 shows projected posterior means from held-out videos into $\mathbb{R}^2$ via UMAP ([McInnes et al., 2018](#)). Insets show four sample frames and their two nearest neighbors in the projected space. The nearest neighbor frames show that points close together in latent space are similar in the physical state of the agent, while freely varying in camera angle, agent color, and background video.

6. Conclusion

Most modern probabilistic methods rely on concurrently trained parametric recognition and generative models. In

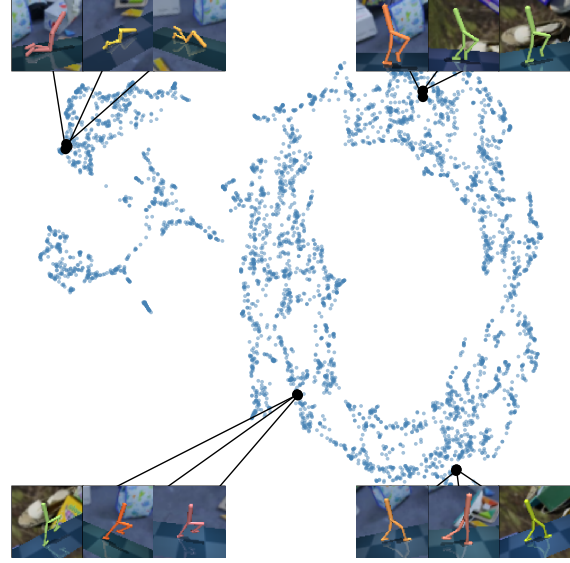


Figure 7. 2-dimensional UMAP projections of RP-GSSM posterior means on held-out data sequences, along with four selected corresponding frames and each of their two nearest neighbors in projected space.

contrast, the RP-GSSM specifies the “generative” model implicitly in terms of the recognition process. Consequently, learning is not driven by a reconstruction objective, but rather by aligning recognition and inference with the specified prior structure, i.e. by finding latent representations that capture dependencies between different observational factors. We rely on the fact that time-series data naturally provides such factors in the form of observations at different time steps, and posit that “good” latent representations are those that account for the temporal structures and correlations in the observed time series. Indeed, our experiments show that this learning objective is useful for producing latent representations that more reliably filter out distractors in the observations, even if these distractors have their own transient internal structure (e.g. natural images and videos).

The general idea behind the RP-GSSM can be extended to more complicated graphical models. One future direction of work is to extend the RP-GSSM to switching dynamical systems ([Dong et al., 2020](#)). Another is to apply the RP-GSSM with control inputs to problems such as simultaneous localization and mapping ([Çatal et al., 2021](#)), optimal control ([Watson et al., 2019](#)), and model-based reinforcement learning ([Levine, 2018](#)). Finally, additional data modalities such as sound, text, or reward can be incorporated into the graphical model. The resulting increase in the number of conditional independence relationships is expected to strengthen the RP-GSSM learning signal, improving its ability to disregard irrelevant features.

Impact Statement

This paper presents work whose goal is to advance the field of machine learning, specifically the recovery of latent structure in time series. The societal consequences of this work are thus unremarkable.

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A. Proof of Lemma 2.1

Lemma 2.1. *Let the GSSM on observations $\mathbf{x} = (x_1, \dots, x_T)$ and latents $\mathbf{z} = (z_1, \dots, z_T)$, with parameters $\Theta = (m_1, Q_1, A, b, Q)$ and an (arbitrary) emission distribution $e(x_t|z_t)$, be stable. Then there exists another stable GSSM with parameters $\tilde{\Theta} = (\tilde{m}_1, \tilde{Q}_1, \tilde{A}, \tilde{b}, \tilde{Q})$ and emission distribution $\tilde{e}$ such that $p_{\Theta, e}(\mathbf{x}) = p_{\tilde{\Theta}, \tilde{e}}(\mathbf{x})$, the stationary distribution of the latent variables (under $\tilde{\Theta}$) is $\mathcal{N}(0, I)$, and $\tilde{b} = 0$.*

Proof. The GSSM with parameters

$$\Theta = (m_1, Q_1, A, b, Q)$$

and emission distribution e defines a joint distribution on $(z_1, \dots, z_T)$ and $(x_1, \dots, x_T)$ according to

$$\begin{aligned} z_1 &\sim \mathcal{N}(m_1, Q_1), \\ z_t|z_{t-1} &\sim \mathcal{N}(Az_{t-1} + b, Q) && \text{for } t = 2, \dots, T, \text{ and} \\ x_t|z_t &\sim e(x_t|z_t) && \text{for } t = 1, \dots, T. \end{aligned}$$

Let G be invertible, and consider the variables $u_t := Gz_t + c$ for all t . Using the properties of Gaussian distributions under affine transformations of variables we have

$$u_1 \sim \mathcal{N}(Gm_1 + c, GQ_1G^\top)$$

and

$$u_t|z_{t-1} \sim \mathcal{N}(GAz_{t-1} + c + Gb, GQG^\top)$$

or, noting that $z_t = G^{-1}(u_t - c)$,

$$u_t|u_{t-1} \sim \mathcal{N}(GAG^{-1}u_{t-1} + (I - GAG^{-1})c + Gb, GQG^\top) \quad \text{for } t = 2, \dots, T.$$

Furthermore,

$$x_t|u_t \sim e(x_t|G^{-1}(u_t - c)),$$

or

$$x_t|u_t \sim e(x_t|h(u_t)),$$

where $h : u_t \mapsto G^{-1}(u_t - c)$.

Therefore, the variables $(u_1, \dots, u_T)$ and emissions $(x_1, \dots, x_T)$ are jointly distributed according to a GSSM with parameters

$$\tilde{\Theta} := (Gm_1 + c, GQ_1G^\top, GAG^{-1}, (I - GAG^{-1})c + Gb, GQG^\top). \quad (\text{S1})$$

and an emission distribution $\tilde{e}(x_t|u_t) = e(x_t|h(u_t))$. The emission variables were left unchanged by the transformation, and so

$$p_{\Theta, e}(\mathbf{x}) = p_{\tilde{\Theta}, \tilde{e}}(\mathbf{x}).$$

The transition matrices on z_t and u_t , given by A and GAG^{-1} respectively, are similar and hence have the same eigenvalues. Therefore $\rho(A) = \rho(GAG^{-1})$ and the GSSM given by $\tilde{\Theta}$ for any invertible G is stable.

Next, let the stationary distribution of z_t be $\mathcal{N}(m_\infty, Q_\infty)$. The stationary distribution p_∞ satisfies

$$p_\infty(z_{t+1}) = \int p(z_{t+1}|z_t)p_\infty(z_t) dz_t,$$

which yields the *Lyapunov equation*:

$$Q_\infty = AQ_\infty A^\top + Q,$$

as well as an equation for the stationary mean:

$$m_\infty = Am_\infty + b. \quad (\text{S2})$$

Denote the stationary distribution of the system with parameters $\tilde{\Theta}$ by $\mathcal{N}(\tilde{m}_\infty, \tilde{Q}_\infty)$. $\tilde{Q}_\infty$ satisfies

$$\tilde{Q}_\infty = GQ_\infty G^\top.$$

Q_∞ is symmetric positive definite and can be decomposed as $Q_\infty = USU^\top$ where U is orthogonal and S is diagonal. Setting $G := S^{-1/2}U^\top$ gives

$$\tilde{Q}_\infty = I.$$

Next, note that in Equation (S2), if the bias b is 0, then $Am_\infty = m_\infty$. However, this would imply that A has an eigenvalue of 1; a contradiction. Therefore, $b = 0 \implies m_\infty = 0$.

The bias of the system defined by $\tilde{\Theta}$, given in Equation (S1), is $(I - GAG^{-1})c + Gb$. Setting to 0 and solving for c ,

$$c = (GAG^{-1} - I)^{-1}Gb.$$

This equation has a unique solution for c if and only if $|GAG^{-1} - I| \neq 0$. This is the case if and only if GAG^{-1} does not have an eigenvalue equal to 1. However, as established previously, GAG^{-1} has eigenvalues equal to those of A , which cannot be 1 by assumption. Therefore, there exists a vector c such that the system defined by $\tilde{\Theta}$ has zero bias, and therefore, as argued above, also has stationary mean $\tilde{m}_\infty$ equal to 0. This completes the proof. $\square$

B. Proof of Theorem 3.1

We begin with proofs of two preliminary results: Lemmas B.1 and B.2.

Lemma B.1. *Under assumptions (iii-iv) of Theorem 3.1, $\tilde{p}_\theta(\mathbf{z}|\mathbf{x})$ is an exponential family with sufficient statistics $t(\mathbf{z})$, natural parameters $\sum_t \eta_t(x_t)$, and base measure $r(\mathbf{z}) := p(\mathbf{z}) / \prod_t F_t(\mathbf{z})$:*

$$\tilde{p}_\theta(\mathbf{z}|\mathbf{x}) = r(\mathbf{z})e^{t(\mathbf{z})^\top (\sum_t \eta_t(x_t)) - \Phi_F(\sum_t \eta_t(x_t))}.$$

$\Omega_F := \text{dom}(\Phi_F)$ is open, convex and connected, Φ_F is real-analytic on Ω_F , and $\nabla^2 \Phi_F(\eta) \succ 0$ for every $\eta \in \Omega_F$.

Proof. We first compute $\tilde{p}_\theta(\mathbf{x})$ by integrating $\mathbf{z}$ out of Equation (5) directly:

$$\begin{aligned} \tilde{p}_\theta(\mathbf{x}) &= \left(\prod_{t=1}^T p_t(x_t) e^{-\Phi(\eta_t(x_t))} \right) \int p(\mathbf{z}) \prod_{t=1}^T \frac{e^{t(\mathbf{z})^\top \eta_t(x_t)}}{F_t(\mathbf{z})} d\mathbf{z} \\ &= \left(\prod_{t=1}^T p_t(x_t) e^{-\Phi(\eta_t(x_t))} \right) e^{\Phi_F(\sum_t \eta_t(x_t))}, \end{aligned}$$

where $\Phi_F(\eta) := \log \int r(\mathbf{z}) e^{t(\mathbf{z})^\top \eta} d\mathbf{z}$. Applying Bayes' rule, $\tilde{p}_\theta(\mathbf{z}|\mathbf{x}) = \tilde{p}_\theta(\mathbf{z}, \mathbf{x}) / \tilde{p}_\theta(\mathbf{x}) = r(\mathbf{z}) e^{t(\mathbf{z})^\top \eta - \Phi_F(\eta)}$, where $\eta = \sum_t \eta_t(x_t)$, giving the stated exponential family. It is a standard property of exponential families that $\Omega_F := \{\eta : \Phi_F(\eta) < \infty\}$ is convex and connected, and that Φ_F is real-analytic on its interior (Wainwright & Jordan, 2008).

Finally, we show that the exponential family is minimal, which will imply the result. Note that by assumption (iii) $p(\mathbf{z})$ is Gaussian and hence positive everywhere on its domain, and $F_t(\mathbf{z})$ is positive and bounded everywhere on its domain, by assumption (iv) of Theorem 3.1. Hence $r(\mathbf{z}) = p(\mathbf{z}) / \prod_t F_t(\mathbf{z})$ is positive everywhere. A positive density cannot concentrate on the zero set of a nonzero polynomial in $\mathbf{z}$, since such a set has Lebesgue measure zero. $t(\mathbf{z})$ is a nonzero polynomial, and hence no nonzero affine combination $a_0 + a^\top t(\mathbf{z})$ vanishes r -almost everywhere, so the exponential family is minimal. By exponential family theory, minimality is equivalent to $\nabla^2 \Phi_F(\eta) = \text{Cov}_\eta(t(\mathbf{z})) \succ 0$ for every $\eta \in \Omega_F$ (Wainwright & Jordan, 2008). $\square$

Lemma B.2. *Under assumptions (i-ii) of Theorem 3.1, if $\tilde{p}_\theta(\mathbf{x}) = \tilde{p}_{\theta'}(\mathbf{x})$, then there exists for every t a diffeomorphism g_t , defined almost everywhere, such that $\eta_t = g_t \circ \eta'_t$.*

Proof. We show that for every t , $\eta_t(x_t)$ is a minimal sufficient statistic for $\mathbf{x}_{-t}$ under $\tilde{p}_\theta(\mathbf{x})$. This implies the existence of g_t .

By definition, $\eta_t(x_t)$ is minimal sufficient for $\mathbf{x}_{-t}$ under $\tilde{p}_\theta(\mathbf{x})$ if and only if, for any x_t, x'_t , the likelihood ratio $\tilde{p}_\theta(x_t, \mathbf{x}_{-t})/\tilde{p}_\theta(x'_t, \mathbf{x}_{-t})$ is independent of $\mathbf{x}_{-t}$ precisely when $\eta_t(x_t) = \eta_t(x'_t)$. By Lemma B.1,

$$\frac{\tilde{p}_\theta(x_t, \mathbf{x}_{-t})}{\tilde{p}_\theta(x'_t, \mathbf{x}_{-t})} = \frac{p_t(x_t)e^{-\Phi(\eta_t)}}{p_t(x'_t)e^{-\Phi(\eta'_t)}} \cdot e^{\Phi_F(\eta_t + \boldsymbol{\eta}_{-t}) - \Phi_F(\eta'_t + \boldsymbol{\eta}_{-t})}, \quad \boldsymbol{\eta}_{-t} := \sum_{s \neq t} \eta_s(x_s),$$

which is independent of $\mathbf{x}_{-t}$ if and only if $\mathbf{x}_{-t} \mapsto \Phi_F(\eta_t + \boldsymbol{\eta}_{-t}) - \Phi_F(\eta'_t + \boldsymbol{\eta}_{-t})$ is constant. By assumption (i), the image U of $\boldsymbol{\eta}_{-t}$ is a nonempty open set. Constancy is equivalent to $\nabla \Phi_F(\eta_t + v) = \nabla \Phi_F(\eta'_t + v)$ for all $v \in U$. Lemma B.1 gives $\nabla^2 \Phi_F \succ 0$ everywhere on Ω_F . Therefore, Φ_F is strictly convex and hence has injective gradient, so for fixed v , $\nabla \Phi_F(\eta_t + v) = \nabla \Phi_F(\eta'_t + v)$ implies $\eta_t + v = \eta'_t + v$, i.e., $\eta_t = \eta'_t$. Combined with the converse (if $\eta_t = \eta'_t$ the ratio is trivially independent of $\mathbf{x}_{-t}$), this proves $\eta_t(x_t)$ is minimal sufficient for $\mathbf{x}_{-t}$.

Now suppose $\tilde{p}_\theta(\mathbf{x}) = \tilde{p}_{\theta'}(\mathbf{x})$. Since η_t and η'_t are both minimal sufficient for $\mathbf{x}_{-t}$ under this common distribution, they carry exactly the same information about $\mathbf{x}_{-t}$: any two minimal sufficient statistics for the same quantity are in one-to-one correspondence (a coarser correspondence would let one statistic distinguish two values of x_t that the other cannot, contradicting minimality of the latter). This gives a bijection $g_t: \eta'_t \mapsto \eta_t$. By assumption (ii), η_t and η'_t are differentiable with almost-everywhere full-rank Jacobians. Hence, g_t is itself almost-everywhere differentiable with invertible Jacobian, i.e., a diffeomorphism almost everywhere. $\square$

Theorem 3.1. *Let $\tilde{p}_\theta$ and $\tilde{p}_{\theta'}$ be two limiting RP-GSSMs with parameters $\theta = (A, Q, \boldsymbol{\eta})$, $\theta' = (A', Q', \boldsymbol{\eta}')$, where $\boldsymbol{\eta}$ denotes a collection of arbitrarily nonstationary recognition factor natural parameters: $(\eta_1, \dots, \eta_T)$. Assume that*

- (i) *Each η_t has open image;*
- (ii) *The η_t are differentiable with full-rank Jacobians almost everywhere;*
- (iii) *Q and Q' are positive definite;*
- (iv) *$\exists L > 0$ such that $\nabla^2 \log F_t(\mathbf{z}) \preceq LI \forall t, \mathbf{z}$;*
- (v) *$T \geq 3$.*

If $\tilde{p}_\theta(\mathbf{x}) = \tilde{p}_{\theta'}(\mathbf{x})$, then there exists an invertible linear mapping relating the models' natural parameters and latent spaces: $\mathbf{z} = G\mathbf{z}'$.

Proof. By Lemma B.2, for every t there exists a diffeomorphism g_t , a.e., with $\eta_t = g_t \circ \eta'_t$; let $J_t(\eta'_t)$ denote its Jacobian. By Lemma B.1 applied to both models,

$$\begin{aligned} \tilde{p}_\theta(\mathbf{x}) &= \left(\prod_{s=1}^T p(x_s) e^{-\Phi(\eta_s(x_s))} \right) e^{\Phi_F(\boldsymbol{\eta})}, \\ \tilde{p}_{\theta'}(\mathbf{x}) &= \left(\prod_{s=1}^T p(x_s) e^{-\Phi(\eta'_s(x_s))} \right) e^{\Phi_{F'}(\boldsymbol{\eta}')}, \end{aligned}$$

where $\boldsymbol{\eta} := \sum_s \eta_s(x_s)$, $\boldsymbol{\eta}' := \sum_s \eta'_s(x_s)$, and $\Phi_{F'}$ is defined as in Lemma B.1 but for model θ' . Equating $\tilde{p}_\theta(\mathbf{x}) = \tilde{p}_{\theta'}(\mathbf{x})$ and taking logarithms,

$$\Phi_F(\boldsymbol{\eta}) - \sum_{s=1}^T \Phi(\eta_s(x_s)) = \Phi_{F'}(\boldsymbol{\eta}') - \sum_{s=1}^T \Phi(\eta'_s(x_s)) \quad (\text{S3})$$

for all $\mathbf{x}$.

Fix t and differentiate Equation (S3) with respect to η'_t . Since $\boldsymbol{\eta} = \sum_s \eta_s(x_s)$ and only $\eta_t = g_t(\eta'_t)$ depends on η'_t (with Jacobian J_t), while $\boldsymbol{\eta}' = \sum_s \eta'_s(x_s)$ depends on η'_t directly (Jacobian I),

$$J_t(\eta'_t)^\top [\nabla \Phi_F(\boldsymbol{\eta}) - \nabla \Phi(\eta_t)] = \nabla \Phi_{F'}(\boldsymbol{\eta}') - \nabla \Phi(\eta'_t). \quad (\dagger_t)$$

Write $\mu_F := \nabla \Phi_F(\eta)$, $\mu_t := \nabla \Phi(\eta_t)$, $\mu_{F'} := \nabla \Phi_{F'}(\eta')$, $\mu'_t := \nabla \Phi(\eta'_t)$. Note that $\mu_F, \mu_{F'}$ do not depend on which index was differentiated, since η, η' themselves do not. Repeating for some $s \neq t$ and subtracting $(\dagger_s) - (\dagger_t)$, the $\mu_{F'}$ terms cancel:

$$(J_s(\eta'_s) - J_t(\eta'_t))^\top \mu_F = J_s(\eta'_s)^\top \mu_s - J_t(\eta'_t)^\top \mu_t - \mu'_s + \mu'_t.$$

Denote the left-hand coefficient $C(\eta'_s, \eta'_t) := J_s(\eta'_s) - J_t(\eta'_t)$ and the right-hand side $b(\eta'_s, \eta'_t)$. Both depend only on η'_s, η'_t , not on the remaining $\eta'_r, r \neq s, t$. Since $T \geq 3$ by assumption (v), fix η'_s, η'_t and let η'_r vary, for a third index $r \notin \{s, t\}$. By assumption (i), η_r , and hence η , ranges over a nonempty open set. Further, $\nabla^2 \Phi_F \succ 0$ everywhere on Ω_F (Lemma B.1), so $\nabla \Phi_F$ is a local diffeomorphism by the inverse function theorem. Hence, $\mu_F = \nabla \Phi_F(\eta)$ also ranges over a nonempty open set as η_r varies, while the relation $C(\eta'_s, \eta'_t)^\top \mu_F = b(\eta'_s, \eta'_t)$ stays fixed. A nonzero linear map cannot send an open set to a single point, so $C(\eta'_s, \eta'_t) = 0$. Substituting back into the identity then forces $b(\eta'_s, \eta'_t) = 0$ as well. Since s, t, η'_s, η'_t were arbitrary, $J_s(\eta'_s) = J_t(\eta'_t)$ holds for every pair and every choice of arguments. Fixing η'_t at any reference value shows $J_s(\eta'_s)$ does not in fact depend on η'_s , so $J_t \equiv \mathcal{G}$ is a single invertible matrix, the same for every t . Integrating the constant Jacobian, $g_t(v) = \mathcal{G}v + c_t$ for constants c_t .

It remains to show that $c_t = 0$ for all t , and that $\mathcal{G}$ corresponds to a linear transformation between z and z' . Substituting $J_t \equiv \mathcal{G}$ into $b = 0$ gives $\mathcal{G}^\top \mu_s(\eta'_s) - \mu'_s(\eta'_s) = \mathcal{G}^\top \mu_t(\eta'_t) - \mu'_t(\eta'_t)$ for all s, t and all η'_s, η'_t independently. Since the two sides are functions of independent arguments, each equals a common constant κ , i.e., $\mathcal{G}^\top \nabla \Phi(\mathcal{G}v + c_t) = \nabla \Phi(v) + \kappa$ for every t, v , equivalently $\Phi(\mathcal{G}v + c_t) = \Phi(v) + \kappa^\top v + d_t$ for constants d_t . Write $v = (h, \Lambda)$ for the natural parameters of the Gaussian family with sufficient statistic $t(z)$, so that $\Phi(h, \Lambda) = -\frac{1}{4}h^\top \Lambda^{-1}h - \frac{1}{2}\log \det(-2\Lambda)$, and let $D := \dim(z)$.

We now show $c_t = 0$ and $\kappa = 0$ for every t , and that $\mathcal{G}$ is a congruence. Fix t and abbreviate $c := c_t, d := d_t$. Write $(\alpha(v), \beta(v)) := \mathcal{G}v + c$ for the two blocks of the (affine) shifted map, so that $\Phi(\alpha(v), \beta(v)) = \Phi(v) + \kappa^\top v + d$ holds for v in an open subset of $\mathbb{R}^D \times \{\Lambda \prec 0\}$. Both sides are real-analytic wherever defined and this domain is connected, so the identity extends to every v with $\beta(v) \prec 0$. First, fix Λ and let h range over $\mathbb{R}^D$: since $\beta(v)$ must stay negative definite for every such h while β is affine in h , its linear-in- h part must vanish (otherwise β is unbounded in some direction of h and eventually fails to be negative definite), so $\beta = \beta(\Lambda)$ depends only on Λ . Second, with β fixed in h , match the h -quadratic terms on both sides at fixed Λ : only $-\frac{1}{4}h^\top \Lambda^{-1}h$ and $-\frac{1}{4}\alpha(v)^\top \beta(\Lambda)^{-1}\alpha(v)$ contribute, since $\kappa^\top v$ is linear in h . Writing $\alpha(v) = Nh + \gamma(\Lambda)$ for the h -linear part N of α , this forces $N^\top \beta(\Lambda)^{-1}N = \Lambda^{-1}$ for every Λ , i.e., $\beta(\Lambda) = N\Lambda N^\top$. Now, β is also affine in Λ with some constant term by definition, while $N\Lambda N^\top$ has none, matching constant terms gives $c_\Lambda = 0$ (writing $c = (c_h, c_\Lambda)$). Third, set $h = 0$: using $\beta(\Lambda) = N\Lambda N^\top$, the identity becomes $-\frac{1}{4}\gamma(\Lambda)^\top (N\Lambda N^\top)^{-1}\gamma(\Lambda) = \langle \kappa_\Lambda, \Lambda \rangle$ for every Λ . Since γ is affine in Λ , the left side is a quadratic form in Λ with negative-definite kernel $(N\Lambda N^\top)^{-1}$, while the right side is linear in Λ . Equality for all Λ forces the quadratic part to vanish identically, hence $\gamma \equiv 0$ (so $c_h = 0$ and $\alpha(v) = Nh$ exactly) and $\kappa_\Lambda = 0$.

With $c = 0, \mathcal{G}v = (Nh, N\Lambda N^\top)$ exactly, for the invertible $D \times D$ matrix N . A direct computation, completing the square as above, gives

$$\Phi(Nh, N\Lambda N^\top) = \Phi(h, \Lambda) - \log|\det N|,$$

with no term linear in v . Comparing against $\Phi(\mathcal{G}v) = \Phi(v) + \kappa^\top v + d$, which must also hold, forces $\kappa = 0$ and $d = -\log|\det N|$. Since t was arbitrary, $c_t = 0$ for every t and $\mathcal{G}$ acts on every $v = (h, \Lambda)$ as $(Nh, N\Lambda N^\top)$ for a single invertible matrix N , shared across t .

This is exactly the natural parameter action of a linear substitution in z -space: for any invertible P , if z' has natural parameters (h', Λ') for sufficient statistic $t(z') = (z', z'z'^\top)$, setting $z := Pz'$ gives, after substituting $z' = P^{-1}z$ into $\exp(h'^\top z' + z'^\top \Lambda' z')$, natural parameters $(P^{-\top}h', P^{-\top}\Lambda'P^{-1})$ for z . This matches $(Nh', N\Lambda'N^\top)$ exactly when $P = N^{-\top}$. Relabeling $N^{-\top}$ as G , this gives $z = Gz'$, completing the proof. $\square$

C. The RP-GSSM Loss

C.1. Variational Gaps

The RP-GSSM free energy, as defined in Equation (8), is given by

$$\mathcal{F}(\{q^n\}_{n=1}^N, \theta) = \sum_{n=1}^N \left[\sum_{t=1}^T \left(\left\langle \log \frac{f_{\phi_t}(z_t | x_t^n)}{F_{\phi_t}(z_t)} \right\rangle_{q^n(z_t)} + \log p_0(x_t^n) \right) - \text{KL}(q^n(z) || p_\eta(z)) \right]. \quad (8)$$

The free energy is a lower bound to the log-likelihood of the data under the RP-GSSM joint distribution (Equation (1)):

$$\mathcal{F}(\{q^n\}_{n=1}^N, \theta) \leq \sum_{n=1}^N \log \tilde{p}_\theta^{\mathbf{X}}(\mathbf{x}^n),$$

where $\mathbf{x}^n = (x_1^n, \dots, x_T^n)$. The difference between the data log-likelihood and the free energy is called a *variational gap* and can be written explicitly:

$$\log \tilde{p}_\theta^{\mathbf{X}}(\mathbf{x}^n) - \mathcal{F}(\{q^n\}_{n=1}^N, \theta) = \sum_{n=1}^N \text{KL}(q^n(z) \parallel \tilde{p}_\theta^{\mathbf{X}}(z | \mathbf{x}^n)).$$

Applying the interior variational bound described in Equation (9), we obtain the RP-GSSM auxiliary free energy:

$$\mathcal{G}(q, \theta) \stackrel{+c}{=} \sum_{t=1}^T \left(\langle \log f_\phi^\Delta(z_t | x_t) \rangle_{q(z_t)} - \log \Gamma_\phi^t \right) - \text{KL}(q(z) \parallel p_\eta(z)), \quad (\text{S4})$$

where the terms $\log \Gamma_{\phi_t}^{tn}$ are given explicitly in Equation (9):

$$\begin{aligned} \Gamma_{\phi_t}^{tn} &= \int F_{\phi_t}(z_t) g_\omega^{tn}(z_t) \mathrm{d}z_t \\ &= \frac{1}{N} \sum_{n'=1}^N \int f_{\phi_t}(z_t | x_t^{n'}) g_\omega^{tn}(z_t) \mathrm{d}z_t. \end{aligned} \quad (\text{S5})$$

The gap induced by using Jensen's inequality in Equation (9) can also be written explicitly:

$$\begin{aligned} \log \tilde{p}_\theta^{\mathbf{X}}(\mathbf{x}^n) - \mathcal{G}(\{q^n\}_{n=1}^N, \{g_\omega^{tn}\}_{n=1, t=1}^{N, T}, \theta) \\ = \sum_{n=1}^N \text{KL}(q^n(z) \parallel \tilde{p}_\theta^{\mathbf{X}}(z | \mathbf{x}^n)) - \sum_{n=1}^N \sum_{t=1}^T \text{KL}\left(q^n(z_t) \parallel \frac{F_{\phi_t}(z_t) g_\omega^{tn}(z_t)}{\Gamma_{\phi_t}^{tn}}\right). \end{aligned}$$

The former variational gap vanishes when q^n approaches the true posterior implied by the RP-GSSM joint. The latter gap vanishes when $g_\omega^{tn}(z_t) \propto q^n(z_t)/F_{\phi_t}(z_t)$, as also implied by Jensen's inequality in Section 3.

C.2. Loss Derivation

The full RP-GSSM auxiliary free energy is provided above in Equation (S4). Although in the main text we assume Gaussian distributions throughout, here we allow any exponential family with constant base measure for generality (dropping parameter subscripts for clarity):

$$\begin{aligned} f(z_t | x_t^n) &= e^{\eta(x_t^n)^\top t(z_t) - \Phi(\eta(x_t^n))} \\ q^n(z_t) &= e^{(\eta_{q_t}^n)^\top t(z_t) - \Phi(\eta_{q_t}^n)} \\ p(z_t) &= e^{\eta_{0t}^\top t(z_t) - \Phi(\eta_{0t})}. \end{aligned}$$

I.e., the terms above belong to the same exponential family but with different natural parameters. Let the auxiliary factors $g^{tn}(z_t)$ have the same shape and their own natural parameters $\tilde{\eta}_t^n$:

$$g^{tn}(z_t) = e^{(\tilde{\eta}_t^n)^\top t(z_t)}.$$

The auxiliary factors do not have to be normalized, so we model them without log-normalizer (Φ) terms. With this parametrization, the Γ^{tn} terms from Equation (S5) become

$$\begin{aligned} \Gamma^{tn} &= \frac{1}{N} \sum_{n'=1}^N \int e^{(\eta(x_t^{n'}) + \tilde{\eta}_t^n)^\top t(z_t) - \Phi(\eta(x_t^{n'}))} \mathrm{d}z_t \\ &= \frac{1}{N} \sum_{n'=1}^N e^{\Phi(\eta(x_t^{n'}) + \tilde{\eta}_t^n) - \Phi(\eta(x_t^{n'}))}. \end{aligned} \quad (\text{S6})$$

In Section 3 we introduced the parametrizations $g^{tn}(z_t) \propto q^n(z_t)/p(z_t)$ and $f(z_t|x_t^n) \propto p(z_t)f^\Delta(z_t|x_t^n)$. In the interior variational bound we multiply and divide by g^{tn} , so these terms can be multiplied by arbitrary constants which cancel out. The terms $f(z_t|x_t^n)$, on the other hand, are normalized to be valid probability distributions. In natural parameter space the equivalent parametrizations are

$$\begin{aligned}\tilde{\eta}_t^n &= \eta_{qt}^n - \eta_{0t} \\ \eta(x_t^n) &= \eta_{0t} + \eta^\Delta(x_t^n),\end{aligned}$$

where $\eta^\Delta(x_t^n)$ are the natural parameter corresponding to $f^\Delta(z_t|x_t^n)$ (which is a member of the same exponential family as all other distributions above). In addition to supporting the convergence $F(z_t) \rightarrow p(z_t)$ in the large-data in-model limit, this parametrization ensures that the natural parameters in Equation (S6), i.e., $\eta(x_t^n)$ and $\eta(x_t^{n'}) + \tilde{\eta}_t^n$, are valid natural parameters for all n, n', t . In particular, when all distributions are Gaussian, the parametrization ensures that all covariance matrices are positive definite, avoiding singularities.

The normalizing constant for $f(z_t|x_t^n)$, denoted Z^{tn} , is given by

$$f(z_t|x_t^n) = \underbrace{e^{-\Phi(\eta_{0t} + \eta^\Delta(x_t^n)) + \Phi(\eta_{0t}) + \Phi(\eta^\Delta(x_t^n))}}_{1/Z^{tn}} p(z_t) f^\Delta(z_t|x_t^n).$$

Putting everything together, the interior variational bound is

$$\begin{aligned}\left\langle \log \frac{f(z_t|x_t^n)}{F(z_t)} \right\rangle_{q^n(z_t)} &\geq \left\langle \log \frac{f(z_t|x_t^n)g^{tn}(z_t)}{q^n(z_t)} \right\rangle_{q^n(z_t)} - \log \Gamma^{tn} \\ &= \langle \log f^\Delta(z_t|x_t^n) \rangle_{q^n(z_t)} - \log Z^{tn} - \log \Gamma^{tn} \\ &= \langle \log f^\Delta(z_t|x_t^n) \rangle_{q^n(z_t)} - \log \tilde{\Gamma}^{tn} + \log N,\end{aligned}$$

where

$$\log \tilde{\Gamma}^{tn} = \Phi(\eta_{0t} + \eta^\Delta(x_t^n)) - \Phi(\eta_{qt}^n) - \Phi(\eta^\Delta(x_t^n)) + \log \sum_{n'=1}^N e^{\Phi(\eta_{qt}^n + \eta^\Delta(x_t^{n'})) - \Phi(\eta_{0t} + \eta^\Delta(x_t^{n'}))}.$$

This yields the final loss presented in Section 3:

$$\mathcal{G}(\{q^n\}_{n=1}^N, \theta) = \sum_{n=1}^N \left[\sum_{t=1}^T \left(\langle \log f_\phi^\Delta(z_t|x_t^n) \rangle_{q^n(z_t)} - \log \tilde{\Gamma}_{\eta, \phi}^{tn} \right) - \text{KL}(q^n(z) || p_\eta(z)) \right],$$

where $\theta = (\eta, \phi)$. The terms g_ω^{tn} are dropped from the arguments of $\mathcal{G}$ because they are now defined in terms of θ and the q^n .

C.3. Augmented Kalman Smoothing

D. Implementation Details

All implementation was done in `jax` (Bradbury et al., 2018), using the `dynamax` package (Chang et al., 2024) for Kalman smoothing.

D.1. Baselines

Here we provide details of the baseline models used in this work. We follow Zhao & Linderman (2023) in the implementation of the SVAE and DKF baselines, and our code for these was adapted from their code repository. These models (except for DKF-NL-NL) have a linear-Gaussian prior chain over the latents and neural-network parametrized Gaussian emissions with constant diagonal covariance:

$$\begin{aligned}z_1 &\sim \mathcal{N}(m_1, Q_1) \\ z_t|z_{t-1} &\sim \mathcal{N}(Az_{t-1} + b, Q) \\ x_t|z_t &\sim \mathcal{N}(\mu_\theta(z_t), \sigma^2 I)\end{aligned}$$

Algorithm 2 KalmanSmoother

Output: Smoothed marginal means μ_t^s and covariances Σ_t^s for $t = 1, \dots, T$
 // Forward Pass: Kalman Filter
 $\mu_{1|0} \leftarrow m_1$
 $\Sigma_{1|0} \leftarrow Q_1$
for $t = 1$ **to** T **do**
 $K_t \leftarrow \Sigma_{t|t-1} C_t^\top (C_t \Sigma_{t|t-1} C_t^\top + R_t)^{-1}$
 $\mu_{t|t} \leftarrow \mu_{t|t-1} + K_t (x_t - (C_t \mu_{t|t-1} + d_t))$
 $\Sigma_{t|t} \leftarrow (I - K_t C_t) \Sigma_{t|t-1}$
 if $t < T$ **then**
 $\mu_{t+1|t} \leftarrow A_t \mu_{t|t} + b_t$
 $\Sigma_{t+1|t} \leftarrow A_t \Sigma_{t|t} A_t^\top + Q_t$
 end if
end for
 // Backward Pass: Rauch-Tung-Striebel Smoother
 $\mu_T^s \leftarrow \mu_{T|T}$
 $\Sigma_T^s \leftarrow \Sigma_{T|T}$
for $t = T - 1$ **down to** 1 **do**
 $J_t \leftarrow \Sigma_{t|t} A_t^\top \Sigma_{t+1|t}^{-1}$
 $\mu_t^s \leftarrow \mu_{t|t} + J_t (\mu_{t+1}^s - \mu_{t+1|t})$
 $\Sigma_t^s \leftarrow \Sigma_{t|t} + J_t (\Sigma_{t+1}^s - \Sigma_{t+1|t}) J_t^\top$
end for
return $\{\mu_t^s, \Sigma_t^s\}_{t=1}^T$

Algorithm 3 AugmentedKalmanSmoother

Input: RP-GSSM parameters $\theta = (A, Q, \phi)$, data $x_{1:T}$
Output: Smoothed marginal means μ_t^s and covariances Σ_t^s for $t = 1, \dots, T$
for $t = 1$ **to** T **do**
 Compute $\mu_\phi(x_t), \Sigma_\phi(x_t)$
end for
 $\{\mu_t^s, \Sigma_t^s\}_{t=1}^T \leftarrow \text{KalmanSmoother}(0, I, \{A, 0, Q, I, 0, \Sigma_\phi(x_t)\}_{t=1}^T, \{\mu_\phi(x_t)\}_{t=1}^T)$
return $\{\mu_t^s, \Sigma_t^s\}_{t=1}^T$

The four DKF baselines differ in their variational posterior families. These are as follows:

DKF-L-MF:

$$q_\phi(z_{1:T}) = \prod_{t=1}^T \mathcal{N}(z_t | \mu_{\phi,t}(x_{1:T}), \Sigma_{\phi,t}(x_{1:T}))$$

where $\{\mu_{\phi,t}(x_{1:T}), \Sigma_{\phi,t}(x_{1:T})\}_{t=1}^T$ are the outputs of a bi-directional RNN.

DKF-L-L:

$$q_\phi(z_{1:T}) = \mathcal{N}(z_t | \mu_{\phi,1}(x_{1:T}), \Sigma_{\phi,1}(x_{1:T})) \prod_{t=2}^T \mathcal{N}(z_t | A z_{t-1} + \mu_{\phi,t}(x_{1:T}), \Sigma_{\phi,t}(x_{1:T}))$$

where $\{\mu_{\phi,t}(x_{1:T}), \Sigma_{\phi,t}(x_{1:T})\}_{t=1}^T$ are the outputs of a bi-directional RNN.

DKF-L-NL:

$$q_\phi(z_{1:T}) = \mathcal{N}(z_1 | \mu_{\phi,1}(x_{1:T}), \Sigma_{\phi,1}(x_{1:T})) \cdot \prod_{t=2}^T \mathcal{N}(z_t | g_\phi(z_{1:t-1}, u_{\phi,t}(x_{1:T})), G_{\phi,t}(z_{1:t-1}, u_{\phi,t}(x_{1:T})))$$

where g_ϕ and G_ϕ are neural networks, and $\{u_{\phi,t}(x_{1:T})\}_{t=1}^T$ are the outputs of a bi-directional RNN.

DKF-L-CL:

$$q_\phi(z_{1:T}) = \mathcal{N}(z_1 | \mu_{\phi,1}(x_{1:T}), \Sigma_{\phi,1}(x_{1:T})) \prod_{t=2}^T \mathcal{N}(z_t | Az_{t-1} + \mu_{\phi,t}(x_{1:T}), \Sigma_{\phi,t}(x_{1:T}))$$

where $\{\mu_{\phi,t}(x_{1:T}), \Sigma_{\phi,t}(x_{1:T})\}_{t=1}^T$ are the outputs of a convolutional neural network applied across the time dimension.

DKF-NL-NL has the same posterior structure as DKF-L-NL, but with prior transitions parametrized by an MLP followed by LayerNorm and Gaussian noise.

The KVAE and CPC models are parametrized in the same way as originally published in [Fraccaro et al. \(2017\)](#) and [van den Oord et al. \(2018\)](#), respectively. In particular, the CPC autoregressive model is parametrized with a forward GRU. The KVAE chain mixture weight RNN is parametrized with a forward LSTM.

D.2. Auxiliary Decoders

The loss used to train the RP-GSSM auxiliary decoder (Figure 4b, 6) was

$$\mathcal{L}(\psi) = \sum_{n=1}^N \sum_{t=1}^T \langle \log p_\psi(x_t^n | z_t) \rangle_{q^n(z_t)},$$

where ψ parametrizes the generative model:

$$p_\psi(x_t^n | z_t) = \mathcal{N}(x_t^n | f_\psi(z_t), \sigma^2 I),$$

for $\psi = (\varphi, \sigma^2)$. The loss was estimated using samples $z_t \sim q^n(z_t)$ and optimized using the reparametrization trick to backpropagate through the samples. This is identical to the generative component of the loss used by the SVAE and the DKF.

The loss used to train the CPC auxiliary decoder was

$$\mathcal{L}(\varphi) = \sum_{n=1}^N \sum_{t=1}^T \|f_\varphi(c_t^n) - x_t^n\|_2^2,$$

where c_t is the *context vector* output by the recurrent network ([van den Oord et al., 2018](#)) and f_φ is the auxiliary decoder network.

All generative networks f_φ are parametrized as three-layer deconvolutional neural networks followed by a sigmoid function to map to $(0, 1)$.

D.3. Hyperparameters

A learning rate of 10^{-3} and batch size of 32 were used for all runs, except for walker experiments, in which a learning rate of 10^{-4} and batch size of 50 were used. All runs swept over recognition covariance parametrized as either diagonal or as a Cholesky factor. The RP-GSSM was further swept over recognition covariance being data-dependent or data-independent. For all generative baseline methods, the generative covariance was taken to be constant, as is typical in the VAE literature. The generative covariance was parametrized as $\nu^2 I$, with ν swept from $\{1, 0.1, 0.001\}$. The Adam optimizer ([Kingma & Ba, 2017](#)) was used for all runs.

To enforce stability of the learned latent dynamical systems we required $\rho(A) < 1$. A sufficient condition is that the largest singular value of A is less than 1. The set of matrices with singular values less than 1 is convex, simplifying constrained optimization ([Buesing et al., 2012](#)). We enforce the constraint by clipping the singular values of A to the range $(0, 1 - \epsilon)$ after each gradient step. We took $\epsilon = 10^{-3}$.

The linear and all pendulum experiments were trained for 10,000 iterations, whereas all tracking experiments were trained for 5,000 iterations.

The following latent dimensionalities were swept over:

Table S1. Optimal hyperparameters for all RP-GSSM experiments from Section 5. $\mathcal{D}_Z$ = RP-GSSM latent dimensionality; CRC = constant recognition covariance; DRC = diagonal recognition covariance.

Dataset	$\mathcal{D}_Z$	CRC	DRC
Linear	–	Yes	Yes
Pendulum	8	Yes	No
CIFAR10-Pendulum	8	Yes	No
Double Pendulum	8	No	No
CIFAR10-Double Pendulum	16	Yes	No
Tracking	16	Yes	No
DAVIS-Tracking	32	Yes	No

- Pendulum: $\{4, 8\}$;
- Double pendulum: $\{8, 16\}$;
- Tracking: $\{8, 16, 32\}$.

For the linear experiments, the latent dimension of all methods was set equal to the problem’s true latent dimension.

Table S1 shows the optimal parameters for the RP-GSSM in different experiments.

D.4. Controlled RP-GSSM

When modeling control inputs, we parametrize the RP-GSSM prior as follows:

$$z_1 \sim \mathcal{N}(0, I),$$

$$z_{t+1} | z_t, u_t \sim \mathcal{N}(A_\psi(a_t)z_t + b_\psi(a_t), Q).$$

The recognition factors remain the same as in the uncontrolled setting.

D.5. Compute Resources

All linear and all pendulum experiments were run on a single RTX5000 GPU. Tracking experiments were run on a single RTX4090 GPU. Walker experiments were run on a single L40S GPU.

E. Experimental Details

Each dataset consists of data sequences of $T = 100$ time steps. The number of sequences N in each dataset is as follows:

- **Linear:** $N = 200$;
- **Pendulum:** $N = 500$;
- **Double Pendulum:** $N = 376$;
- **Tracking:** $N = 500$;
- **Walker:** $N = 400$.

The **Linear** dataset was sampled from a linear GSSM:

$$z_1 \sim \mathcal{N}(0, I),$$

$$z_t \sim \mathcal{N}(Bz_{t-1}, I - BB^\top),$$

$$x_t \sim \mathcal{N}(Cz_t + d, R).$$

B was a scaled rotation matrix, applying a rotation of $\pi/5$ around a randomly chosen axis in $\mathbb{R}^{D_z}$, with operator norm 0.95; C and d had components sampled i.i.d from $\mathcal{N}(0, 1)$, and $R = 0.3I$. All recognition and generative networks were taken to be single-layer linear.

The **Pendulum** dataset consists of sequences of 24×24 pixel frames sampled at 0.01s intervals. Pixel intensities are normalized to $(0, 1)$ and uncorrelated pixel-level zero-mean Gaussian noise with standard deviation 0.05 is applied to all frames. The **Double Pendulum** observations are 50×50 pixel frames of a real-life double pendulum system, sampled at 0.0075s intervals. Frames are single-channel grayscale for both tasks.

For the **Tracking** dataset, the observation frames comprise 64×64 pixels with three color channels. Each 100 frame-long sequence is equivalent to approximately 5s of real time. The “head”-direction (represented by an arrow in the observations) is computed using a low-pass filter on the velocity vector.

The **Walker** dataset was taken from the DeepMind Control Vision Benchmark (DMC-VB; Ortiz et al. (2024)) with the walker-walk environment, dynamic distractors, and expert policy. The first 100 timesteps of each data sequence were used. Observation frames consist of 64×64 with three color channels.

F. Further Results

F.1. Empirical Validation of $F_{\phi_t}(z_t) \approx p_\eta(z_t)$

In Figure S1 we show quantitative and qualitative evidence supporting the approximation $F_{\phi_t}(z_t) \approx p_\eta(z_t)$. Note that $F_{\phi_t}(z_t)$ is always a mixture of Gaussians. In Figure S1, we train the RP-GSSM on batches of 32 data sequences for 100 and 25 training iterations on the linear and pendulum tasks, respectively. Then, a new batch of sequences is drawn and the process is repeated. We maintain the entire buffer of training sequences to compute the mixture $F_{\phi_t}(z_t)$.

In Figure S1 (left) we plot the normalized error of the first three centered moments of $F_{\phi_t}(z_t)$ (computed in closed form) relative to those of the Gaussian $p_\eta(z_t)$, averaged over t , when trained on a ground-truth linear dataset with latent and observation dimensionality both equal to 2. The errors of the first three centered moments decrease during training.

Figure S1 (center) measures the error between $F_{\phi_t}(z_t)$ and $p_\eta(z_t)$ on the pendulum task under two metrics: maximum mean discrepancy (MMD), and Fisher divergence. For the MMD, we use a Gaussian RBF kernel $k(x, y) = \exp(-\|x - y\|_2^2 / 2K)$, where the bandwidth is taken to be equal to the latent dimension K , a common heuristic for Gaussian data. The MMD between distributions p and q can be expressed as

$$\text{MMD}^2(p, q) = \mathbb{E}_{x, y \sim p} k(x, y) + \mathbb{E}_{x, y \sim q} k(x, y) - 2\mathbb{E}_{x \sim p, y \sim q} k(x, y)$$

and can be computed in closed form in the case of (mixtures of) Gaussian distributions. We approximate Fisher divergence by moment-matching $F_{\phi_t}(z_t)$ to a single Gaussian $\mathcal{N}(\hat{\mu}_t, \hat{\Sigma}_t)$. $p_\eta(z_t)$ is set up so that it equals $\mathcal{N}(0, I)$ for all t , so that we can compute the Fisher divergence

$$D_{\mathcal{F}}(\mathcal{N}(\hat{\mu}_t, \hat{\Sigma}_t) \parallel \mathcal{N}(0, I)) = \|\hat{\mu}_t\|_2^2 + \text{Tr}(\hat{\Sigma}_t) + \text{Tr}(\hat{\Sigma}_t^{-1}) - 2K.$$

Both metrics start near 0, as the recognition factors learn to have uninformative, very large, covariances early on in training. This changes over the first ~ 1000 iterations of training, upon which we see that $F_{\phi_t}(z_t)$ begins to approach the prior.

To visualize how the mixture of Gaussian $F_{\phi_t}(z_t)$ can approach a single Gaussian, we train the RP-GSSM on the pendulum task with a latent dimensionality of 1. Once the buffer reached 3200 training sequences, we compute each individual recognition factor as well as $F_{\phi_t}(z_t)$. In Figure S1 (right) we plot the individual recognition factors’ densities for all t and n , as well as their average over t and n (which we denote $\bar{F}$). There is diversity in the locations and shapes of the recognition factors, but $\bar{F}$ very closely matches the true standard Gaussian prior.

F.2. Data Efficiency

The RP-GSSM is the only method that is able to infer angular velocity when trained on only 64 data sequences (Figure S2).

F.3. Additional Regression Results

Figure S3 is identical to Figure 3, but using kernel ridge regression instead of linear regression. Regression was performed with an RBF kernel and with a fixed regularization constant across all models and tasks.

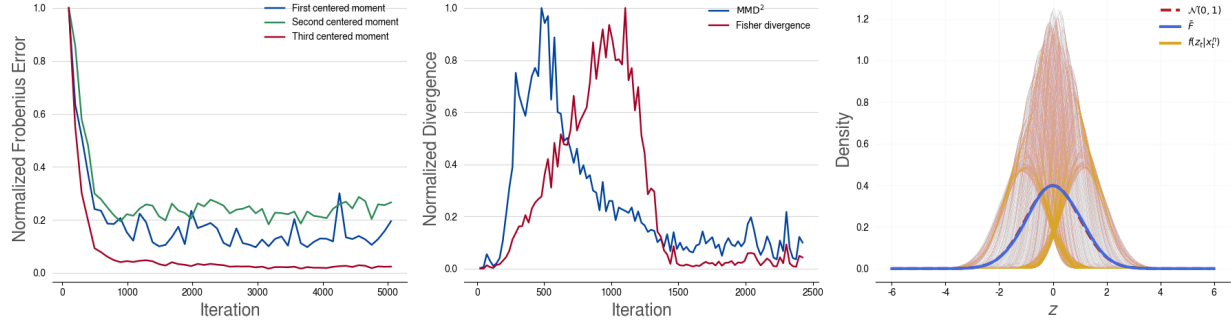


Figure S1. **Left:** The normalized Frobenius norm over the course of training on a 2-dimensional linear task between the first three centered moments of $F_{\phi_t}(z_t)$, and those of $p_\eta(z_t)$, averaged over t . **Center:** The normalized MMD and approximate Fisher divergences over the course of training on the pendulum task between $F_{\phi_t}(z_t)$ and $p_\eta(z_t)$. **Right:** Individual recognition factors over t and $N = 3200$ sequences, as well as their corresponding $F_{\phi_t}(z_t)$ and the true $p_\eta(z_t) = \mathcal{N}(0, 1)$.

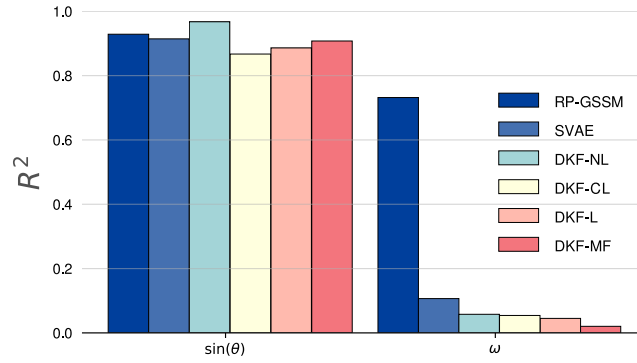


Figure S2. Linear regression R^2 scores to sine of pendulum angle, $\sin \theta$, and angular velocity, ω , of all methods trained on the first 64 sequences of the single-pendulum dataset.

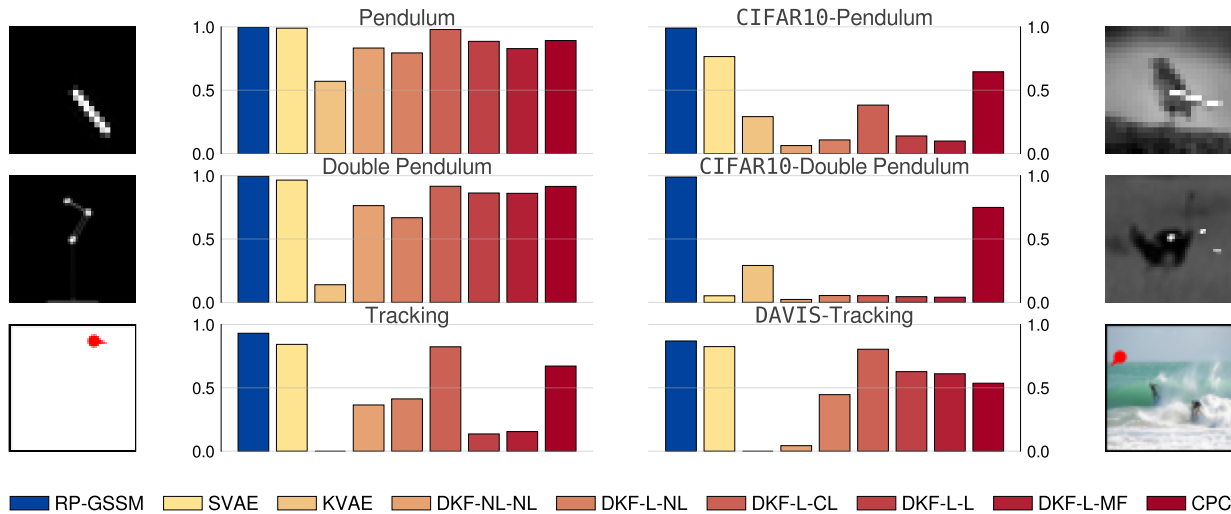


Figure S3. Mean kernel ridge regression R^2 scores.

Figures S4 and S5 show R^2 scores from linear and kernel ridge regression, respectively, to all target variables. The target variables are as follows:

- **Top two rows:** sine of angle, $\sin \theta$, and angular velocity, ω ;
- **Middle two rows:** sine of primary angle with respect to the origin, $\sin \theta_1$, and sine of secondary angle with respect to the primary angle, $\sin \theta_2$;
- **Bottom two rows:** rat position, x , rat velocity, v , and rat head direction, x_{head} . These have dimensionality 2, 2, and 1, respectively.

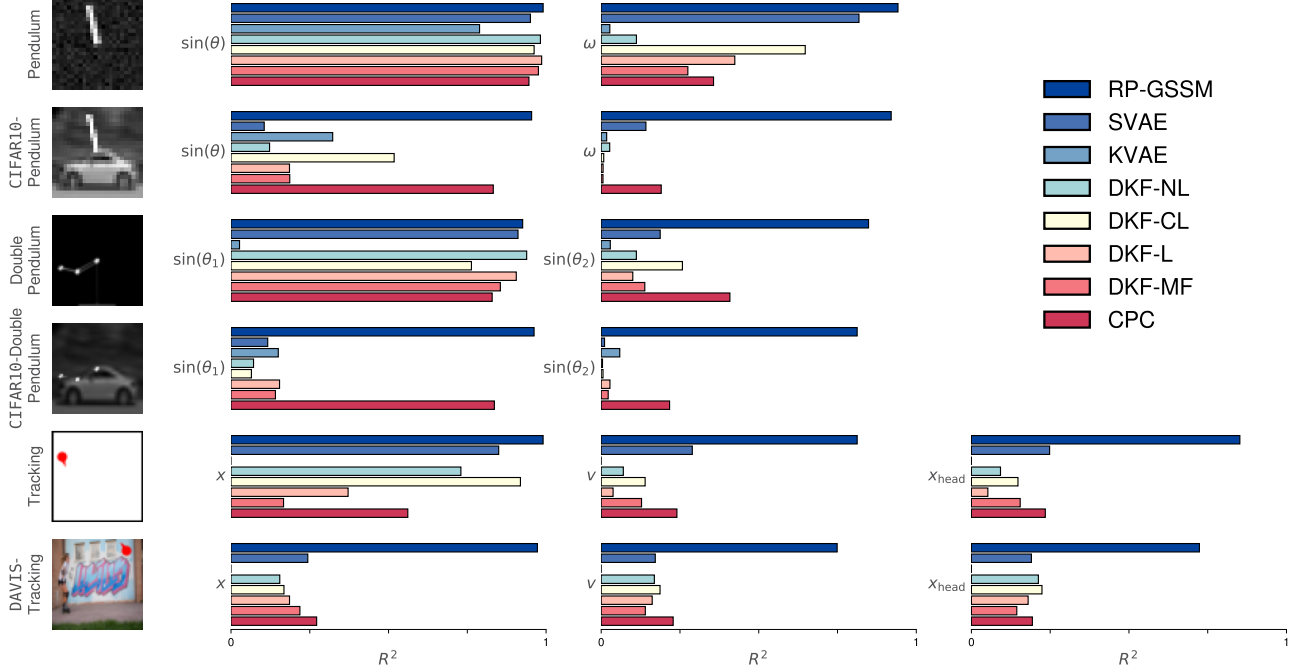


Figure S4. Linear regression R^2 scores to all target variables.

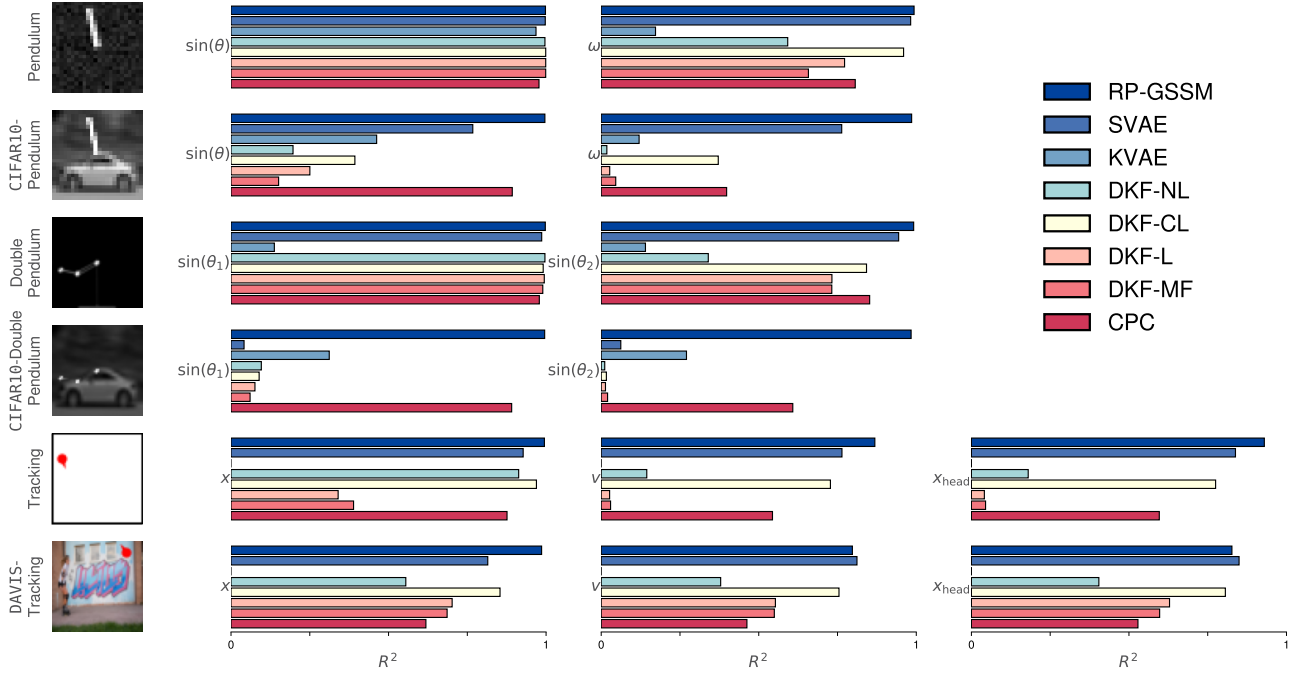


Figure S5. Kernel ridge regression R^2 scores to all target variables.

F.4. Computational Complexity and Runtimes

Figure S6 shows the mean time it takes for each model to complete one iteration, averaged over all runs in the sweeps, for the three linear tasks. The RP-GSSM has the fastest iterations. However, these times are implementation-dependent and may not necessarily be representative of true computational complexity. For completeness, we normalize each model's times relative to its mean time on the smallest linear task. Normalized mean times are shown in Figure S7; the RP-GSSM times scale with latent and observation dimensionalities at a similar rate to the baseline models.

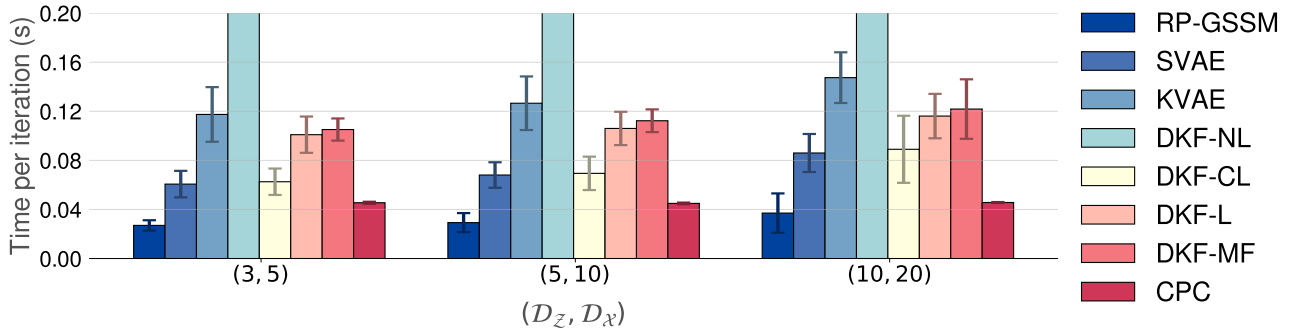


Figure S6. Average times per iteration across the full sweeps for the linear tasks. Error bars indicate standard deviation. The times for DKF-NL (approximately 0.3s) are significantly higher than for other models, so they are cropped for visual clarity.

F.5. Robustness Results for DAVIS-Tracking

Here we visualize the inferred latent states for a single trajectory with 5 different background videos (Figure S8). The results demonstrate the RP-GSSM's robustness to changes in background distractor—something the generative and contrastive methods do not share.

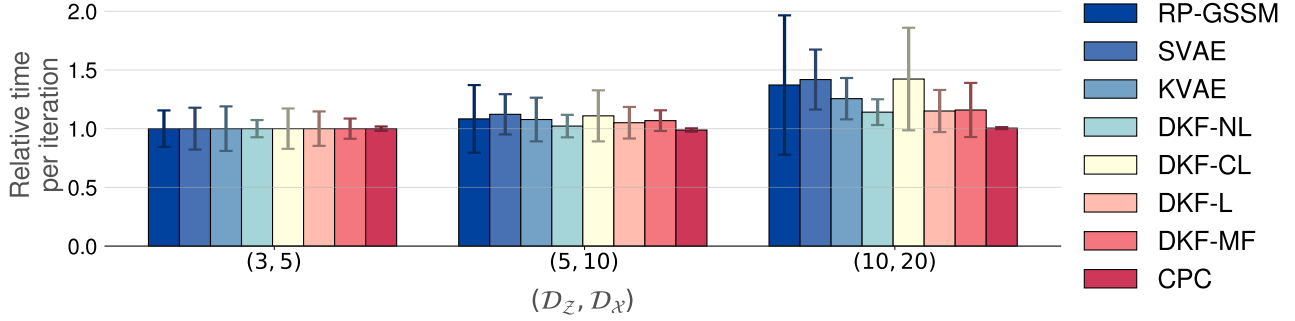


Figure S7. Average times per iteration, normalized relative to the mean time per iteration for the smallest linear task, across the full sweeps for the linear tasks. Error bars indicate standard deviation.

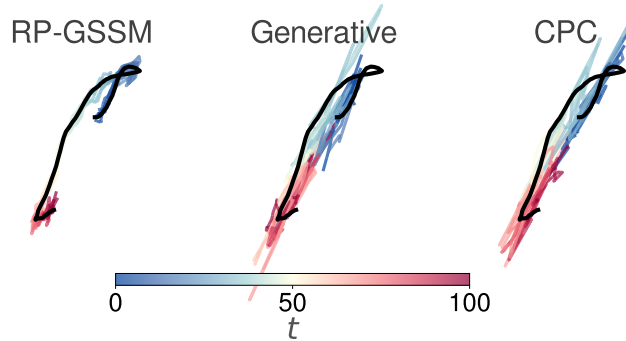


Figure S8. The RP-GSSM performs more accurate tracking in the DAVIS-Tracking dataset than the best generative and CPC baselines. The black line is a sample ground truth trajectory and the colored lines are inferred trajectories across five different background videos.

F.6. Walker Results and Baseline

Here we report additional results from the Walker experiment (Section 5.5) and compare them to the world model of the Dreamer algorithm (Hafner et al., 2020) - a generative latent variable model with non-linear dynamics. We plot the R^2 scores from kernel ridge regression between inferred posterior means and ground truth states on the training batch of every 100th iteration in Figure S9. RP-GSSM has a significantly higher R^2 score than the Dreamer world model.

In Figure S10 we directly map the inferred posterior means and sampled action-conditioned rollouts from the trajectory in Figure 6 with neural networks. We again compare the results from RP-GSSM to Dreamer with 32-dimensional and 8-dimensional latent spaces. In both instances, the inferred and predicted latents from RP-GSSM map much closer to the ground truth states than the Dreamer world model.

In Figure S12 we show reconstructions of Dreamer with 32-dimensional latents, as well as generated images from an auxiliary decoder trained to map the Dreamer latents to clean observations, on the same validation sequence as in Figure 6. While the Dreamer reconstructions of the distracted observations may appear cleaner to the eye, the mapping to the clean observations suggests that this is due to the Dreamer latents containing more information about the distractions than RP-GSSM. Dreamer is not able to track or predict the agent configuration as closely as RP-GSSM, as indicated in Figure S9.

We also show the results from the auxiliary decoders of the RP-GSSM with 32-dimensional latents in Figures S11 and S12. Additionally, we plot the reconstructions generated by the Dreamer generative model on the distracted observations, as well as from an auxiliary decoder trained to map the inferred latents from the distracted observations to the clean observations. Here, we see clearly that the RP-GSSM captures the dynamics of the agent well, while not encoding information about distractors. Dreamer, on the other hand, contains a lot of information about the distractors, leading to poor performance.

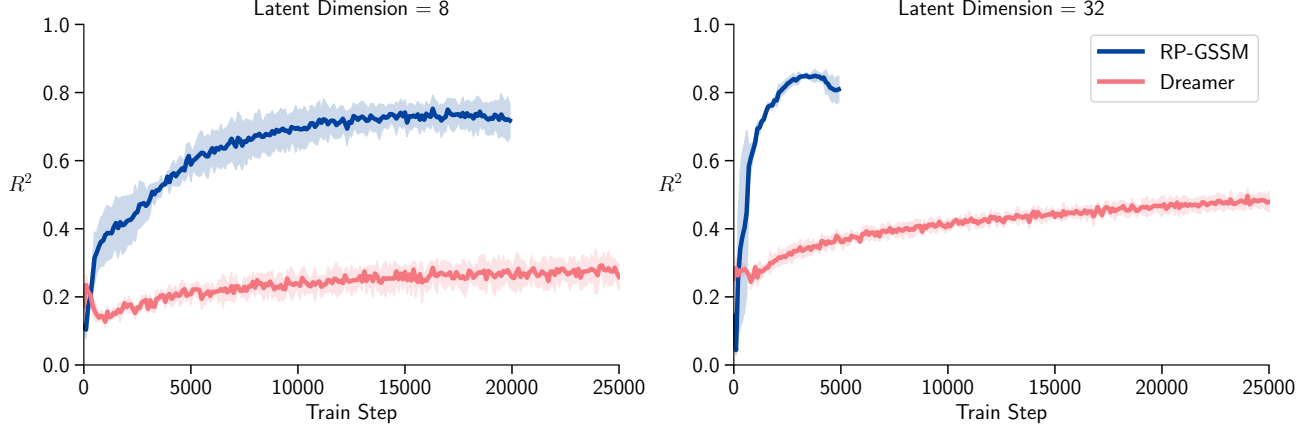


Figure S9. Kernel ridge regression R^2 scores between inferred posterior means and ground truth latent states on the training batch from the Walker dataset, logged every 100 training iterations (Section 5.5). We compare RP-GSSM to Dreamer at 2 different sizes of latent dimension, plotting the mean and standard deviation over 3 random seeds. RP-GSSM significantly outperforms Dreamer in both cases, and takes far fewer training iterations to achieve a high R^2 score.

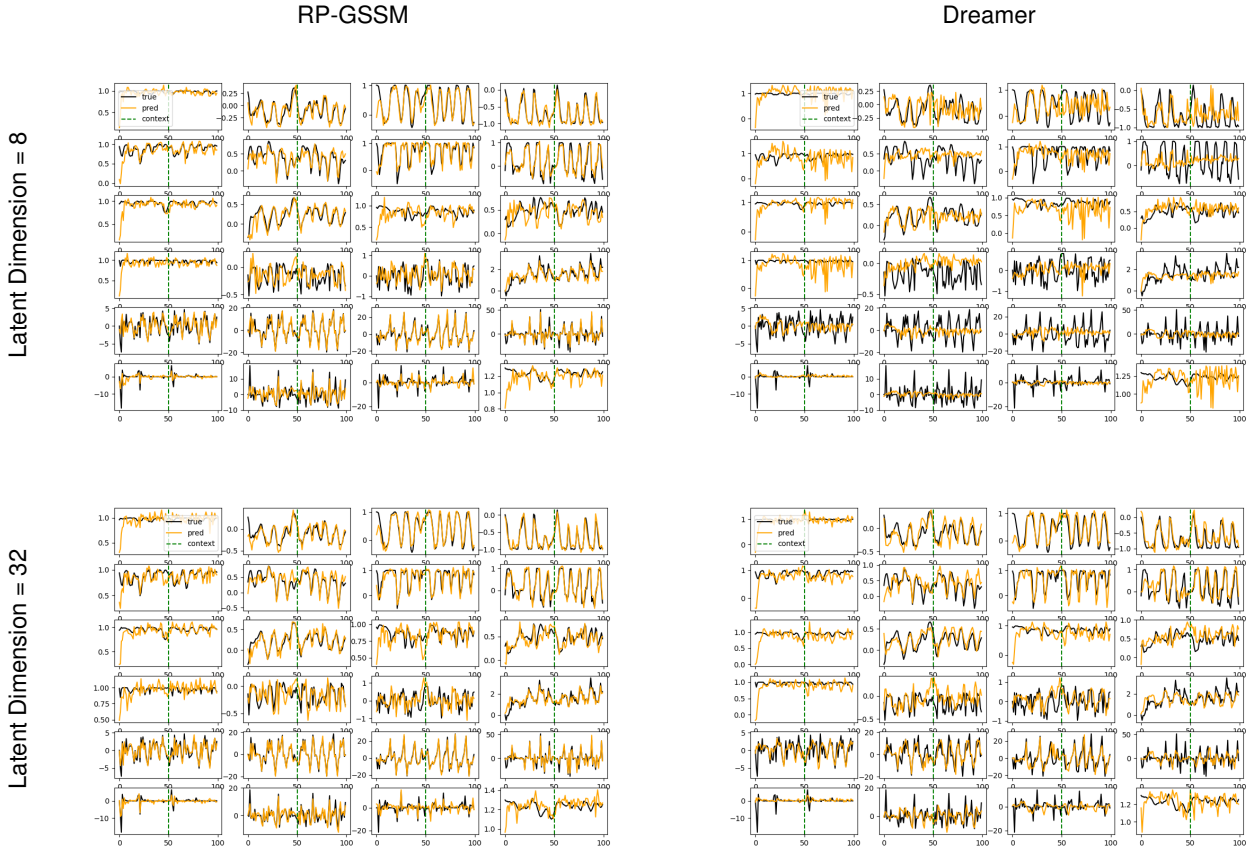


Figure S10. We train a neural network to map from inferred posterior means to ground truth latent states. Here we show the results of mapping from posterior means (for $t < T_{\text{context}}$) and a sampled action-conditioned rollout (for $t \geq T_{\text{context}}$) to the ground truth states for the same hold-out trajectory reconstructed in Figure 6. Each plot represents one dimension of the ground-truth state. **Top:** latent dimension of the models is set to 8. **Bottom:** latent dimension of the models is set to 32. The black lines are the ground-truth states, and the orange are the inferred/predicted latents. The green dashed line is at T_{context} .

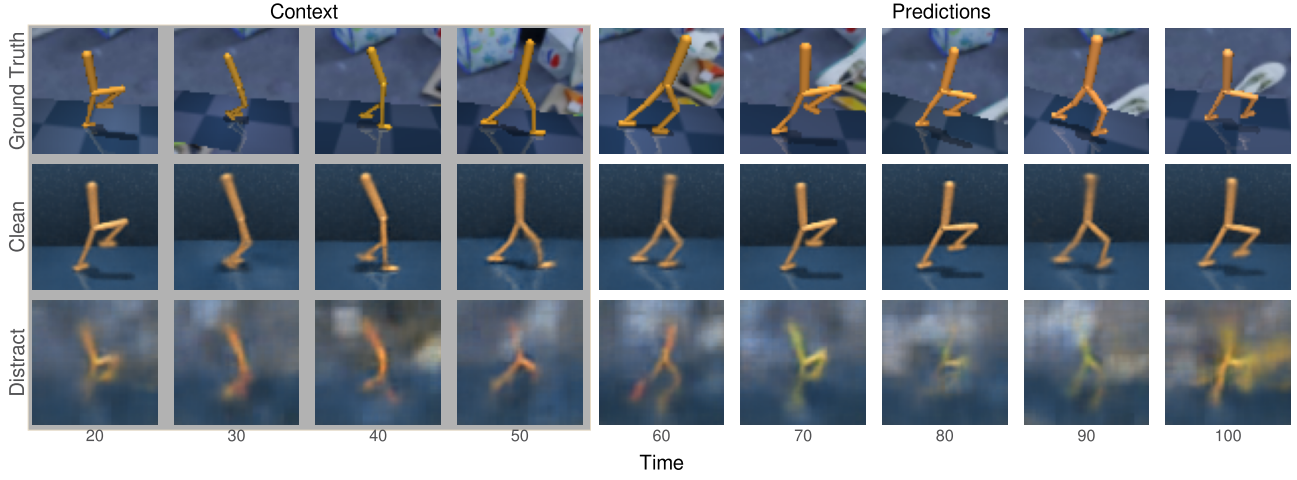


Figure S11. Reconstructions from the RP-GSSM auxiliary decoders with 32-dimensional latent space to validation distract data (“Distract”) as well as to clean observations (“Clean”).

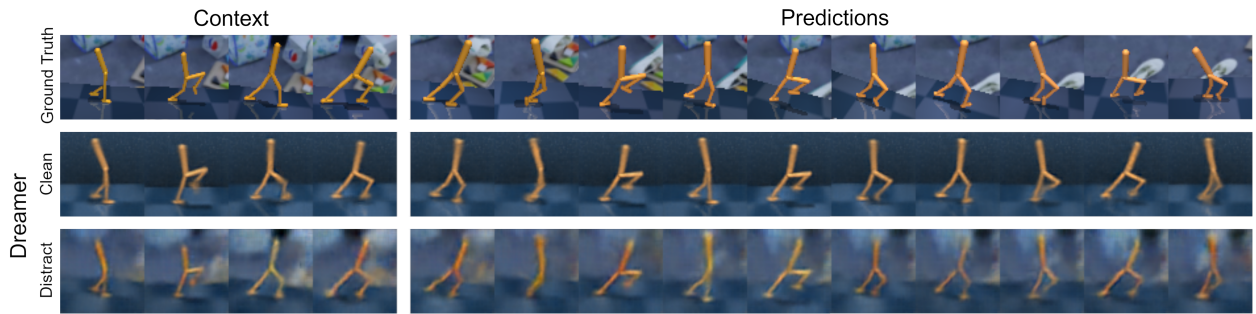


Figure S12. Reconstructions from Dreamer with 32-dimensional latent space (“Distract”) as well as from an auxiliary decoder trained to map Dreamer latents to clean observations (“Clean”).

G. Licenses

We release the RP-GSSM model and code under the CC BY-NC 4.0 license.

The assets used in this work have the following licenses:

- The `CIFAR10` dataset: MIT license;
- The `DAVIS` dataset: BSD license;
- The code for the SVAE and DKF models ([Zhao & Linderman, 2023](#)): MIT license;
- The KVAE model: unknown license;
- The CPC model: unknown license;
- The `simple-dreamer` repository: MIT license;
- The DMC-VB dataset: CC-BY 4.0 license.